

Joint thick-point content for correlated log-correlated fields

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Abstract

We develop a *joint thick-point content* theory for several correlated linear observables of a log-correlated Gaussian field. Our main result is a fully explicit theorem for a concrete star-scale invariant vector field: after the natural renormalization, the joint high-excursion content measure converges, in the weak topology of measures, to the Gaussian multiplicative chaos (GMC) associated with the unique minimum-energy direction selected by the linear constraints, throughout the full subcritical regime $I < 2d$. In the L^2 -phase $I < d$, we moreover recover the exact Hausdorff dimension of the corresponding simultaneous thick-point set on the nontriviality event of the rooted chaos, and in particular the almost sure upper Hausdorff bound. We further explain how this result fits into a broader transfer framework for general log-correlated fields and discuss links with the circular Gaussian free field and with circular ensembles.

Contents

1	Introduction and goals	2
2	A concrete star-scale invariant vector log-field	3
2.1	The Gaussian field model and a concrete construction	3
2.2	GMC for vector fields	6
3	Simultaneous thick points and the minimum-energy direction	7
3.1	Multiple linear observables	7
3.2	The constrained minimum-energy direction	7
4	Joint thick-point content and the main limit theorem	7
4.1	Joint high-excursion content	7
4.2	A precise Gaussian tail constant	8
4.3	Statement of the main theorem (fully proved in L^2 -phase)	10
4.4	Proof of Theorem 4.4	10
5	The exact Hausdorff dimension in the L^2-phase	20
5.1	Lower bound via the rooted chaos	21
5.2	A uniform oscillation estimate on a regular spatial grid	26
5.3	A Gaussian vector thick-point upper bound	29

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6	Full subcritical regime $I < 2d$	32
6.1	A moment input for subcritical GMC	32
6.2	Exact conditional expectation of the thick-point content	32
6.3	Controlling the J_T -error on average	35
6.4	A mesoscopic decoupling and a p -moment fluctuation estimate	38
6.5	Convergence in probability for all $I < 2d$	43
6.6	Remarks	44
7	Transfer issues and links with circle models and circular ensembles	45
8	Discussion and outlook	45
9	Bibliographic note on scalar subcritical GMC input	46

1 Introduction and goals

Gaussian multiplicative chaos (GMC) is the canonical way to exponentiate log-correlated Gaussian fields into random fractal measures. Since Kahane’s seminal work, the subject has developed into a broad interface between probability, statistical mechanics, random geometry, and random matrix theory; see for instance [1, 3, 16]. A particularly influential theme is the relation between GMC and the *thick points* of the underlying field. Recent work has gone well beyond geometric upper and lower bounds for thick-point sets: suitably renormalized thick-point contents can in fact converge to the chaos measure itself; see [8].

The present paper is devoted to the corresponding problem for *several correlated observables at once*. Given a log-correlated Gaussian vector field and finitely many linear functionals, we ask: what is the correct renormalized content of the set of points where all these observables are simultaneously thick? Does this content converge to a canonical random measure, and if so, how is the limit determined by the constraint?

The answer turns out to be both geometric and variational. The joint thick-point content is controlled by a unique *minimum-energy direction* u_* , obtained by solving a convex optimization problem under the given linear constraints. The limiting object is then precisely the GMC associated with u_* , multiplied by an explicit deterministic density.

Positioning of the paper

Conceptually, the paper should be read as a two-level contribution.

- (1) **Core theorem.** We prove, in a concrete star-scale invariant vector model, a fully explicit joint thick-point content theorem throughout the full subcritical range $I < 2d$. Apart from standard scalar GMC input that we quote explicitly, the genuinely joint probabilistic part of the argument is written in a form that can be checked line by line in the explicit model.
- (2) **Broader context.** We then discuss how the mechanism of the theorem may extend to more general log-correlated fields and how it interfaces with circle/GFF and circular-ensemble questions. These parts are included to explain the natural scope of the result, but they are not used as formal inputs in the core proof.

What is actually new here

The novelty of the paper is not a new one-parameter multifractal exponent. Rather, the new point is that a *joint* high-excursion event for several correlated observables gives rise to a canonical measure-valued limit, and that this limit is identified by a clean minimum-energy principle. This is the multi-observable analogue of the now classical “thick points generate chaos” philosophy. At the level of applications, the paper provides a framework in which one can formulate simultaneous thick-point questions for circular ensembles, in the spirit of [7, 9].

Main outputs

The mathematically complete results proved in this paper are:

- a full-subcritical joint thick-point content theorem in an explicit star-scale invariant vector model;
- the exact Hausdorff dimension of the associated simultaneous thick-point set in the L^2 -phase on the nontriviality event of the rooted chaos (and the corresponding almost sure upper bound);
- a detailed discussion of how the theorem may plausibly transfer to more general log-correlated fields and how it connects to circle/GFF and circular-ensemble questions.

How to read the paper

Sections 2–6 contain the core probabilistic argument in the explicit model. The subsequent discussion section explains possible connections of the result with more general log-correlated fields and with circle/random-matrix questions. The final section records the bibliographic note for the scalar subcritical GMC input used in the non- L^2 regime.

2 A concrete star-scale invariant vector log-field

2.1 The Gaussian field model and a concrete construction

Fix $d \geq 1$ and work on the unit cube $D = [0, 1]^d$. Fix an integer $m \geq 1$ and a symmetric positive definite matrix $\Sigma \in \mathbb{R}^{m \times m}$. We endow $\mathbb{R}^m$ with the Σ -inner product

$$\langle u, v \rangle_\Sigma := u^\top \Sigma v, \quad \|u\|_\Sigma^2 := \langle u, u \rangle_\Sigma.$$

Because all later arguments require a *single* process $(X_t)_{t \geq 0}$ with independent increments in the log-scale variable t , we give an explicit white-noise construction.

Definition 2.1 (A compactly supported scale kernel). Fix a function $\varphi \in C_c^\infty(\mathbb{R}^d)$ supported in $B(0, \frac{1}{4})$ and normalized by $\int_{\mathbb{R}^d} \varphi(z)^2 dz = 1$. Define

$$k(w) := \int_{\mathbb{R}^d} \varphi(z) \varphi(z + w) dz.$$

Then k is bounded, uniformly continuous, satisfies $k(0) = 1$, and is compactly supported in $B(0, \frac{1}{2})$.

Definition 2.2 (Star-scale invariant cut-off field (white noise construction)). Let W be an m -dimensional Gaussian white noise on $\mathbb{R}^d \times (0, 1]$ with covariance Σ , meaning that for test functions $F, G \in L^2(\mathbb{R}^d \times (0, 1]; \mathbb{R}^m)$,

$$\mathbb{E}\left[\left(\int F \cdot dW\right)\left(\int G \cdot dW\right)\right] = \int_{\mathbb{R}^d \times (0, 1]} F(z, r)^\top \Sigma G(z, r) \, dz \, dr.$$

For $t \geq 0$ and $x \in \mathbb{R}^d$ define

$$X_t(x) := \int_{e^{-t}}^1 \int_{\mathbb{R}^d} r^{-(d+1)/2} \varphi\left(\frac{x-z}{r}\right) dW(z, r). \quad (1)$$

This is a well-defined $\mathbb{R}^m$ -valued centered Gaussian process in $(t, x) \in [0, \infty) \times \mathbb{R}^d$. All chaos measures will be constructed by restricting the resulting random field to the domain $D = [0, 1]^d$, but in the oscillation and comparison arguments below we will occasionally use the natural extension of X_t to all of $\mathbb{R}^d$.

Remark 2.3 (Underlying probability space). Throughout the paper we realize the white noise W on its canonical Gaussian probability space (e.g. the Borel space of generalized functions on $\mathbb{R}^d \times (0, 1]$ endowed with the corresponding Gaussian law). In particular, the underlying probability space is standard Borel. This will be used when invoking regular conditional probabilities in Lemmas 6.8 and 6.2; the σ -fields that appear there are generated by measurable functionals of W .

Lemma 2.4 (Covariance and independent increments). *For $t, s \geq 0$ and $x, y \in \mathbb{R}^d$,*

$$\mathbb{E}[X_t(x)X_s(y)^\top] = \Sigma K_{t \wedge s}(x, y), \quad K_u(x, y) := \int_{e^{-u}}^1 k\left(\frac{x-y}{r}\right) \frac{dr}{r}. \quad (2)$$

In particular, $\text{Var}(u \cdot X_t(x)) = \|u\|_\Sigma^2 t$. Moreover, for $0 \leq s < t$, the increment field $X_t - X_s$ is independent of the sigma-field generated by $\{X_r : 0 \leq r \leq s\}$.

Proof. By linearity of the stochastic integral and the white-noise covariance,

$$\begin{aligned} \mathbb{E}[X_t(x)X_s(y)^\top] &= \int_{\mathbb{R}^d \times (0, 1]} \left(\mathbb{1}_{[e^{-t}, 1]}(r) r^{-(d+1)/2} \varphi\left(\frac{x-z}{r}\right) \right) \left(\mathbb{1}_{[e^{-s}, 1]}(r) r^{-(d+1)/2} \varphi\left(\frac{y-z}{r}\right) \right)^\top dz \, dr \, \Sigma \\ &= \Sigma \int_{e^{-(t \wedge s)}}^1 \int_{\mathbb{R}^d} r^{-(d+1)} \varphi\left(\frac{x-z}{r}\right) \varphi\left(\frac{y-z}{r}\right) dz \, dr. \end{aligned}$$

Make the change of variables $u = (x - z)/r$, so $z = x - ru$ and $dz = r^d du$. The inner integral becomes

$$\int_{\mathbb{R}^d} r^{-(d+1)} \varphi(u) \varphi\left(u + \frac{x-y}{r}\right) r^d du = \frac{1}{r} \int_{\mathbb{R}^d} \varphi(u) \varphi\left(u + \frac{x-y}{r}\right) du = \frac{1}{r} k\left(\frac{x-y}{r}\right).$$

Substituting gives (2). Setting $x = y$ gives $K_u(x, x) = \int_{e^{-u}}^1 k(0) dr/r = u$, so $\text{Var}(u \cdot X_t(x)) = \|u\|_\Sigma^2 t$.

Finally, the increment $X_t - X_s$ is the same stochastic integral as in (1) but restricted to $r \in (e^{-t}, e^{-s})$. The field generated by $\{X_r : r \leq s\}$ depends only on the white noise on scales $r' \in [e^{-s}, 1]$. These space-scale regions are disjoint up to the boundary slice $r = e^{-s}$, which has zero control measure. Hence, by independent scattering of white noise, the increment is independent of the past. $\square$

Lemma 2.5 (Jointly measurable continuous modification in the scale parameter). *For every $T > 0$, the Gaussian field $(t, x) \mapsto X_t(x)$ on $[0, T] \times D$ admits a jointly measurable version whose sample paths are jointly continuous in (t, x) . Consequently, for every linear map $T_0 : \mathbb{R}^m \rightarrow \mathbb{R}^k$, the field*

$$Y_t(x) := T_0 X_t(x), \quad (t, x) \in [0, \infty) \times D,$$

also admits a jointly measurable version that is jointly continuous on every compact set $[0, T] \times D$. Throughout the paper we work with such a version.

Proof. Fix $T > 0$. We prove a Kolmogorov bound on the parameter set $[0, T] \times D$, viewed as a subset of $\mathbb{R}^{d+1}$.

For $0 \leq s, t \leq T$ and $x, y \in D$, write

$$X_t(x) - X_s(y) = (X_t(x) - X_s(x)) + (X_s(x) - X_s(y)).$$

By Lemma 2.4,

$$\mathbb{E}|X_t(x) - X_s(x)|^2 \leq C_\Sigma |t - s|.$$

It remains to bound the spatial increment at fixed scale $s \in [0, T]$. By the covariance formula (2),

$$\mathbb{E}|X_s(x) - X_s(y)|^2 = 2 \operatorname{Tr}(\Sigma) \int_{e^{-s}}^1 \left(k(0) - k\left(\frac{x-y}{r}\right) \right) \frac{dr}{r}.$$

Since $s \leq T$, we have $r \geq e^{-T}$. Because $k \in C_c^\infty(\mathbb{R}^d)$, it is globally Lipschitz:

$$|k(u) - k(v)| \leq L_k |u - v| \quad (u, v \in \mathbb{R}^d).$$

Hence

$$0 \leq k(0) - k\left(\frac{x-y}{r}\right) \leq L_k \frac{|x-y|}{r}.$$

Therefore

$$\mathbb{E}|X_s(x) - X_s(y)|^2 \leq 2 \operatorname{Tr}(\Sigma) L_k |x - y| \int_{e^{-T}}^1 \frac{dr}{r^2} \leq C_T |x - y|.$$

Combining the time and space bounds yields

$$\mathbb{E}|X_t(x) - X_s(y)|^2 \leq C_T (|t - s| + |x - y|) \quad \text{for all } (t, x), (s, y) \in [0, T] \times D.$$

Since X is Gaussian, all moments are controlled by the second moment. Thus for every $p \geq 1$,

$$\mathbb{E}|X_t(x) - X_s(y)|^p \leq C_{p,T} (|t - s| + |x - y|)^{p/2}.$$

Choose $p > 2(d+1)$. Then $p/2 > d+1$, so Kolmogorov–Chentsov on the $(d+1)$ -dimensional parameter set $[0, T] \times D$ gives a version of $(t, x) \mapsto X_t(x)$ that is jointly continuous on $[0, T] \times D$, hence in particular jointly measurable. Applying the linear map T_0 yields the same property for $Y_t(x) = T_0 X_t(x)$.

To obtain a global version on $[0, \infty) \times D$, let $\tilde{X}^{(N)}$ be the jointly continuous version on $[0, N] \times D$. For each pair $M < N$, the processes $\tilde{X}^{(M)}$ and $\tilde{X}^{(N)}$ are versions of the same Gaussian field on $[0, M] \times D$, so they agree almost surely at every point of a fixed countable dense subset of $[0, M] \times D$. Intersecting over all pairs $M < N$ and all points of the dense subset gives a full-probability event on which $\tilde{X}^{(M)} = \tilde{X}^{(N)}$ on the dense subset, and hence everywhere on $[0, M] \times D$ by joint continuity. We may therefore define a single jointly measurable version $\tilde{X}$ on $[0, \infty) \times D$ by patching these restrictions together. The same construction applies to $Y_t(x) = T_0 X_t(x)$. $\square$

Remark 2.6 (Log-correlation). Because k is compactly supported and $k(0) = 1$, as $t \rightarrow \infty$ one has

$$K_t(x, y) = \log \frac{1}{|x - y|} + O(1) \quad (x \neq y)$$

with the $O(1)$ bounded uniformly over $x, y \in D$. Thus each scalar projection $u \cdot X_t$ is a log-correlated Gaussian field with variance growth $\text{Var}(u \cdot X_t(x)) = \|u\|_\Sigma^2 t$.

2.2 GMC for vector fields

Definition 2.7 (Cut-off GMC). Let $u \in \mathbb{R}^m$ and $t \geq 0$. Define the random Radon measure

$$\mu_t^u(\mathrm{d}x) := \exp\left(u \cdot X_t(x) - \frac{1}{2} \text{Var}(u \cdot X_t(x))\right) \mathrm{d}x = \exp\left(u \cdot X_t(x) - \frac{1}{2} \|u\|_\Sigma^2 t\right) \mathrm{d}x.$$

Lemma 2.8 (Martingale property). *Let*

$$\mathcal{F}_t^X := \sigma(X_r(x) : 0 \leq r \leq t, x \in D).$$

For each Borel set $A \subset D$, the process $(\mu_t^u(A))_{t \geq 0}$ is a nonnegative martingale with respect to the filtration $(\mathcal{F}_t^X)_{t \geq 0}$.

Proof. Fix $0 \leq s < t$. By construction,

$$X_t = X_s + (X_t - X_s),$$

where the increment field $(X_t - X_s)$ is centered Gaussian and independent of the whole past sigma-field $\mathcal{F}_s^X$ by Lemma 2.4. Since X_s is $\mathcal{F}_s^X$ -measurable, for each $x \in D$,

$$\mathbb{E}\left[e^{u \cdot X_t(x) - \frac{1}{2} \|u\|_\Sigma^2 t} \middle| \mathcal{F}_s^X\right] = e^{u \cdot X_s(x) - \frac{1}{2} \|u\|_\Sigma^2 s} \mathbb{E}\left[e^{u \cdot (X_t(x) - X_s(x)) - \frac{1}{2} \|u\|_\Sigma^2 (t-s)}\right].$$

The remaining expectation equals 1, because $u \cdot (X_t(x) - X_s(x))$ is centered Gaussian with variance $\|u\|_\Sigma^2 (t - s)$. Hence

$$\mathbb{E}\left[e^{u \cdot X_t(x) - \frac{1}{2} \|u\|_\Sigma^2 t} \middle| \mathcal{F}_s^X\right] = e^{u \cdot X_s(x) - \frac{1}{2} \|u\|_\Sigma^2 s}.$$

Since the integrand is nonnegative, conditional Fubini yields

$$\mathbb{E}[\mu_t^u(A) \mid \mathcal{F}_s^X] = \mu_s^u(A).$$

This proves the martingale property. $\square$

Theorem 2.9 (Subcritical GMC limit in the L^2 phase (quoted input)). *If $\|u\|_\Sigma^2 < d$, then there exists a non-trivial random Borel measure μ^u on D such that for every bounded continuous $f : D \rightarrow \mathbb{R}$,*

$$\int_D f \, \mathrm{d}\mu_t^u \xrightarrow[t \rightarrow \infty]{L^2, \text{ a.s.}} \int_D f \, \mathrm{d}\mu^u.$$

Proof. This is standard scalar GMC theory in the exact-scaling / star-scale invariant setting considered here; see for instance Kahane [1], Robert–Vargas [2], Rhodes–Vargas [3], and Shamov [5]. In particular, for the scalar chaos martingale $(\mu_t^u)_{t \geq 0}$, one has a non-trivial random measure limit whenever $\|u\|_\Sigma^2 < 2d$, and in the L^2 -phase $\|u\|_\Sigma^2 < d$ the convergence holds in L^2 as well as almost surely. We use this only as quoted background. $\square$

3 Simultaneous thick points and the minimum-energy direction

3.1 Multiple linear observables

Fix integers $k \geq 1$ and nonzero vectors $t_1, \dots, t_k \in \mathbb{R}^m$. Let

$$T := \begin{pmatrix} t_1^\top \\ \vdots \\ t_k^\top \end{pmatrix} \in \mathbb{R}^{k \times m}, \quad Q := T \Sigma T^\top \in \mathbb{R}^{k \times k}.$$

Since Σ is positive definite and the t_i are fixed, Q is symmetric positive semidefinite; we assume throughout that Q is positive definite (equivalently, (t_i) are linearly independent in the Σ -inner product).

Define the k -vector field of observables

$$Y_t(x) := T X_t(x) \in \mathbb{R}^k.$$

Then for each x , $Y_t(x) \sim \mathcal{N}(0, tQ)$.

3.2 The constrained minimum-energy direction

Fix a target slope vector $c = (c_1, \dots, c_k) \in \mathbb{R}^k$. Define

$$\lambda := Q^{-1}c \in \mathbb{R}^k, \quad u_* := T^\top \lambda \in \mathbb{R}^m.$$

Then $T \Sigma u_* = Q \lambda = c$ and

$$\|u_*\|_\Sigma^2 = u_*^\top \Sigma u_* = c^\top Q^{-1}c. \quad (3)$$

We write

$$I := \|u_*\|_\Sigma^2.$$

Lemma 3.1 (Variational characterization). *Among all $u \in \mathbb{R}^m$ satisfying the linear constraints $T \Sigma u = c$, the vector $u_* = T^\top Q^{-1}c$ is the unique minimizer of $\|u\|_\Sigma^2$.*

Proof. Let u satisfy $T \Sigma u = c$. Write $u = u_* + v$. Then $T \Sigma v = 0$. Compute

$$\|u\|_\Sigma^2 = (u_* + v)^\top \Sigma (u_* + v) = \|u_*\|_\Sigma^2 + 2\langle u_*, v \rangle_\Sigma + \|v\|_\Sigma^2.$$

But $\langle u_*, v \rangle_\Sigma = u_*^\top \Sigma v = \lambda^\top T \Sigma v = \lambda^\top 0 = 0$. Hence $\|u\|_\Sigma^2 = \|u_*\|_\Sigma^2 + \|v\|_\Sigma^2 \geq \|u_*\|_\Sigma^2$, with equality iff $v = 0$. $\square$

4 Joint thick-point content and the main limit theorem

4.1 Joint high-excursion content

Let $g : D \rightarrow \mathbb{R}^k$ be a bounded continuous function. Define the joint high-excursion set at scale t :

$$\mathcal{H}_t := \left\{ x \in D : Y_t(x) \geq c t + g(x) \right\},$$

where the inequality is coordinatewise.

Definition 4.1 (Joint thick-point content measure). Define a random measure on D by

$$\nu_t(dx) := t^{k/2} e^{\frac{1}{2} I t} \mathbf{1}_{\mathcal{H}_t}(x) dx. \quad (4)$$

4.2 A precise Gaussian tail constant

The constant in the limit is the orthant-tail prefactor.

Proposition 4.2 (Multivariate orthant tail asymptotics with drift). *Let Q be positive definite and $c \in \mathbb{R}^k$. Assume*

$$\lambda := Q^{-1}c \in (0, \infty)^k \quad (5)$$

(componentwise strict positivity). Let $Z_t \sim \mathcal{N}(0, tQ)$ and let $a \in \mathbb{R}^k$ be fixed. Then, as $t \rightarrow \infty$,

$$\mathbb{P}(Z_t \geq ct + a) = \frac{e^{-\frac{1}{2}c^\top Q^{-1}ct} e^{-\lambda \cdot a}}{(2\pi)^{k/2} t^{k/2} \sqrt{\det Q}} \prod_{i=1}^k \frac{1}{\lambda_i} (1 + o(1)). \quad (6)$$

Moreover, the $o(1)$ is uniform when a ranges over any fixed compact set.

Proof. Write the Gaussian density of Z_t :

$$\phi_t(z) = \frac{1}{(2\pi t)^{k/2} \sqrt{\det Q}} \exp\left(-\frac{1}{2}z^\top (tQ)^{-1}z\right).$$

We evaluate $\mathbb{P}(Z_t \geq ct + a) = \int_{[0, \infty)^k} \phi_t(ct + a + y) dy$. Compute the exponent:

$$\begin{aligned} (ct + a + y)^\top (tQ)^{-1} (ct + a + y) &= \frac{1}{t} (ct + a + y)^\top Q^{-1} (ct + a + y) \\ &= t c^\top Q^{-1} c + 2 c^\top Q^{-1} (a + y) + \frac{1}{t} (a + y)^\top Q^{-1} (a + y). \end{aligned}$$

Hence

$$\begin{aligned} \phi_t(ct + a + y) &= \frac{1}{(2\pi t)^{k/2} \sqrt{\det Q}} \exp\left(-\frac{1}{2}c^\top Q^{-1}ct\right) \exp\left(-c^\top Q^{-1}(a + y)\right) \exp\left(-\frac{1}{2t}(a + y)^\top Q^{-1}(a + y)\right) \\ &= \frac{e^{-\frac{1}{2}c^\top Q^{-1}ct} e^{-\lambda \cdot a}}{(2\pi t)^{k/2} \sqrt{\det Q}} \exp(-\lambda \cdot y) \exp\left(-\frac{1}{2t}(a + y)^\top Q^{-1}(a + y)\right). \end{aligned}$$

Therefore

$$\mathbb{P}(Z_t \geq ct + a) = \frac{e^{-\frac{1}{2}c^\top Q^{-1}ct} e^{-\lambda \cdot a}}{(2\pi t)^{k/2} \sqrt{\det Q}} \int_{[0, \infty)^k} \exp(-\lambda \cdot y) \exp\left(-\frac{1}{2t}(a + y)^\top Q^{-1}(a + y)\right) dy. \quad (7)$$

Fix a compact set $K \subset \mathbb{R}^k$ for a . Since Q^{-1} is positive definite, the quadratic form

$$(a + y)^\top Q^{-1} (a + y)$$

is nonnegative for every $a \in K$ and $y \in [0, \infty)^k$. Therefore the extra exponential factor in (7) is bounded by 1, and the integrand is dominated by

$$e^{-\lambda \cdot y} \exp\left(-\frac{1}{2t}(a + y)^\top Q^{-1}(a + y)\right) \leq e^{-\lambda \cdot y}, \quad t > 0,$$

which is integrable on $[0, \infty)^k$ because $\lambda \in (0, \infty)^k$. Moreover,

$$\int_{[0, \infty)^k} \exp(-\lambda \cdot y) dy = \prod_{i=1}^k \frac{1}{\lambda_i}.$$

To justify the *uniform* convergence in $a \in K$, note that for every $u \geq 0$ we have

$$0 \leq 1 - e^{-u} \leq u.$$

Therefore, for every $a \in K$ and $y \in [0, \infty)^k$,

$$\left| e^{-\frac{1}{2t}(a+y)^\top Q^{-1}(a+y)} - 1 \right| \leq \frac{1}{2t}(a+y)^\top Q^{-1}(a+y) \leq \frac{C'_K}{t}(1 + |y|^2),$$

for some constant $C'_K > 0$ depending only on K and Q . Multiplying by $e^{-\lambda \cdot y}$ and taking the supremum over $a \in K$, we obtain

$$\sup_{a \in K} e^{-\lambda \cdot y} \left| e^{-\frac{1}{2t}(a+y)^\top Q^{-1}(a+y)} - 1 \right| \leq \frac{C'_K}{t} e^{-\lambda \cdot y} (1 + |y|^2).$$

The function $y \mapsto e^{-\lambda \cdot y} (1 + |y|^2)$ is integrable on $[0, \infty)^k$, so integrating the last inequality yields

$$\sup_{a \in K} \left| \int_{[0, \infty)^k} e^{-\lambda \cdot y} \exp\left(-\frac{1}{2t}(a+y)^\top Q^{-1}(a+y)\right) dy - \int_{[0, \infty)^k} e^{-\lambda \cdot y} dy \right| \leq \frac{C''_K}{t} \xrightarrow{t \rightarrow \infty} 0.$$

Thus the integral in (7) converges uniformly in $a \in K$ to $\prod_i 1/\lambda_i$. Substituting this into (7) yields (6). $\square$

Lemma 4.3 (Exact Gaussian orthant factorization). *Assume the hypotheses of Proposition 4.2, and let $Z_\tau \sim \mathcal{N}(0, \tau Q)$ with $\tau > 0$. Then for every $a \in \mathbb{R}^k$ one has the exact identity*

$$\tau^{k/2} e^{\frac{1}{2} I \tau} \mathbb{P}(Z_\tau \geq c\tau + a) = C_{c,Q} e^{-\lambda \cdot a} J_\tau(a), \quad (8)$$

where $I = c^\top Q^{-1} c$, $C_{c,Q}$ is defined in (12), and

$$J_\tau(a) := \frac{\int_{[0, \infty)^k} e^{-\lambda \cdot y} \exp\left(-\frac{(a+y)^\top Q^{-1}(a+y)}{2\tau}\right) dy}{\int_{[0, \infty)^k} e^{-\lambda \cdot y} dy}. \quad (9)$$

Moreover,

$$0 < J_\tau(a) \leq 1 \quad \text{for all } \tau > 0, a \in \mathbb{R}^k.$$

Proof. The proof is just a rewriting of the computation in (7). Namely,

$$\mathbb{P}(Z_\tau \geq c\tau + a) = \frac{e^{-\frac{1}{2} I \tau} e^{-\lambda \cdot a}}{(2\pi\tau)^{k/2} \sqrt{\det Q}} \int_{[0, \infty)^k} e^{-\lambda \cdot y} \exp\left(-\frac{(a+y)^\top Q^{-1}(a+y)}{2\tau}\right) dy.$$

Multiply both sides by $\tau^{k/2} e^{I\tau/2}$. Since

$$C_{c,Q} = \frac{1}{(2\pi)^{k/2} \sqrt{\det Q}} \prod_{i=1}^k \frac{1}{\lambda_i} = \frac{1}{(2\pi)^{k/2} \sqrt{\det Q}} \int_{[0, \infty)^k} e^{-\lambda \cdot y} dy,$$

we obtain (8). The inequality $0 < J_\tau(a) \leq 1$ is immediate from (9). $\square$

4.3 Statement of the main theorem (fully proved in L^2 -phase)

Theorem 4.4 (Joint thick-point content $\Rightarrow$ GMC in the most likely direction). *Assume Q is positive definite and $\lambda = Q^{-1}c \in (0, \infty)^k$. Let $u_* = T^\top Q^{-1}c$ and $I = \|u_*\|_\Sigma^2$. Assume the L^2 -phase condition*

$$I < d. \quad (10)$$

Let $g : D \rightarrow \mathbb{R}^k$ be bounded continuous and define ν_t by (4). Then, as $t \rightarrow \infty$, for every bounded continuous test function $f : D \rightarrow \mathbb{R}$,

$$\nu_t(f) := \int_D f \, d\nu_t \xrightarrow[t \rightarrow \infty]{L^2} C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \mu^{u_*}(\mathrm{d}x), \quad (11)$$

where μ^{u_} is the L^2 GMC limit from Theorem 2.9 and*

$$C_{c,Q} := \frac{1}{(2\pi)^{k/2} \sqrt{\det Q}} \prod_{i=1}^k \frac{1}{\lambda_i}. \quad (12)$$

In particular,

$$\nu_t \xrightarrow[t \rightarrow \infty]{P} C_{c,Q} e^{-\lambda \cdot g} \mu^{u_*} \quad \text{in the weak topology of finite measures on } D. \quad (13)$$

4.4 Proof of Theorem 4.4

The proof is by L^2 identification: we show that for each test function f , $\int f \, d\nu_t$ and $C_{c,Q} \int f e^{-\lambda \cdot g} \mathrm{d}\mu_t^{u_*}$ have the same asymptotic first and second moments, and then use L^2 convergence of $\mu_t^{u_*}$.

Lemma 4.5 (One-point asymptotic under the field). *For each $x \in D$,*

$$\mathbb{E}[\nu_t(\mathrm{d}x)] = t^{k/2} e^{\frac{1}{2}It} \mathbb{P}(Y_t(x) \geq ct + g(x)) \mathrm{d}x \xrightarrow[t \rightarrow \infty]{} C_{c,Q} e^{-\lambda \cdot g(x)} \mathrm{d}x, \quad (14)$$

and the convergence is uniform in $x \in D$.

Proof. Fix x . Since $Y_t(x) \sim \mathcal{N}(0, tQ)$, Proposition 4.2 applied with $a = g(x)$ yields

$$\mathbb{P}(Y_t(x) \geq ct + g(x)) = \frac{e^{-\frac{1}{2}It} e^{-\lambda \cdot g(x)}}{(2\pi)^{k/2} t^{k/2} \sqrt{\det Q}} \prod_{i=1}^k \frac{1}{\lambda_i} (1 + o(1)).$$

Multiplying by $t^{k/2} e^{\frac{1}{2}It}$ yields (14). Uniformity over x holds because D is compact, g is continuous, hence $g(D)$ is a compact subset of $\mathbb{R}^k$, and Proposition 4.2 is uniform for a in compact sets. $\square$

Lemma 4.6 (Two-point asymptotic). *For $x \neq y$ in D , define the limiting covariance kernel*

$$K_\infty^{u_*}(x, y) := \lim_{t \rightarrow \infty} \mathrm{Cov}(u_* \cdot X_t(x), u_* \cdot X_t(y)).$$

Then, as $t \rightarrow \infty$,

$$t^k e^{It} \mathbb{P}(Y_t(x) \geq ct + g(x), Y_t(y) \geq ct + g(y)) \rightarrow C_{c,Q}^2 e^{-\lambda \cdot (g(x) + g(y))} \exp(K_\infty^{u_*}(x, y)), \quad (15)$$

Proof. Fix $x \neq y$ and set

$$r := |x - y|, \quad \rho := \min\{r/2, 1/2\} \in (0, 1/2].$$

For all sufficiently large t we have $e^{-t} < \rho$. Split the field at scale ρ :

$$X_t(z) = X^{(\text{co})}(z) + X_t^{(\text{fi})}(z),$$

where

$$X^{(\text{co})}(z) := \int_{\rho}^1 \int_{\mathbb{R}^d} s^{-(d+1)/2} \varphi\left(\frac{z-w}{s}\right) dW(w, s),$$

and

$$X_t^{(\text{fi})}(z) := \int_{e^{-t}}^{\rho} \int_{\mathbb{R}^d} s^{-(d+1)/2} \varphi\left(\frac{z-w}{s}\right) dW(w, s).$$

Since the scale intervals (e^{-t}, ρ) and $(\rho, 1)$ are disjoint up to the boundary scale $r = \rho$, which has zero control measure, the coarse and fine fields are independent.

We first show that the fine fields at x and y are independent. By Lemma 2.4,

$$\mathbb{E}[X_t^{(\text{fi})}(x)X_t^{(\text{fi})}(y)^{\top}] = \Sigma \int_{e^{-t}}^{\rho} k\left(\frac{x-y}{s}\right) \frac{ds}{s}.$$

For $s \leq \rho$ we have

$$\frac{|x-y|}{s} = \frac{r}{s} \geq \frac{r}{\rho} \geq 2.$$

Since k is supported in $B(0, \frac{1}{2})$, the integrand vanishes identically, hence the covariance is 0. Because the pair is jointly Gaussian, $X_t^{(\text{fi})}(x)$ and $X_t^{(\text{fi})}(y)$ are independent. Applying T , the same holds for the fine observables

$$Y_t^{(\text{fi})}(x) := TX_t^{(\text{fi})}(x), \quad Y_t^{(\text{fi})}(y) := TX_t^{(\text{fi})}(y).$$

Now define

$$A_x := TX^{(\text{co})}(x), \quad A_y := TX^{(\text{co})}(y).$$

Then, conditional on (A_x, A_y) , the vectors

$$Y_t(x) - A_x, \quad Y_t(y) - A_y$$

are independent centered Gaussian vectors with common covariance matrix

$$\tau Q, \quad \tau := t - \log \frac{1}{\rho}.$$

Indeed, by Lemma 2.4,

$$\text{Cov}(A_x) = Q \int_{\rho}^1 \frac{ds}{s} = Q \log \frac{1}{\rho},$$

so the remaining variance is precisely

$$tQ - Q \log \frac{1}{\rho} = \tau Q.$$

Hence, conditional on (A_x, A_y) ,

$$\begin{aligned} & \mathbb{P}(Y_t(x) \geq ct + g(x), Y_t(y) \geq ct + g(y) \mid A_x, A_y) \\ &= \mathbb{P}(Z_\tau \geq c\tau + a_x) \mathbb{P}(Z_\tau \geq c\tau + a_y), \end{aligned}$$

where $Z_\tau \sim \mathcal{N}(0, \tau Q)$ and

$$a_x := c \log \frac{1}{\rho} + g(x) - A_x, \quad a_y := c \log \frac{1}{\rho} + g(y) - A_y.$$

For every deterministic $a \in \mathbb{R}^k$ and $\tau > 0$, the exact tail factorization from the proof of Proposition 4.2 gives

$$\mathbb{P}(Z_\tau \geq c\tau + a) = C_{c,Q} \tau^{-k/2} e^{-\frac{1}{2}I\tau} e^{-\lambda \cdot a} J_\tau(a), \quad (16)$$

where

$$J_\tau(a) := \frac{\int_{[0,\infty)^k} e^{-\lambda \cdot y} \exp\left(-\frac{1}{2\tau}(a+y)^\top Q^{-1}(a+y)\right) dy}{\int_{[0,\infty)^k} e^{-\lambda \cdot y} dy}.$$

For all a, τ we have $0 < J_\tau(a) \leq 1$, and for each fixed a ,

$$J_\tau(a) \xrightarrow{\tau \rightarrow \infty} 1.$$

Applying (16) to a_x and a_y and multiplying, we obtain

$$\mathbb{P}(\cdots \mid A_x, A_y) = C_{c,Q}^2 \tau^{-k} e^{-I\tau} e^{-\lambda \cdot (a_x + a_y)} J_\tau(a_x) J_\tau(a_y).$$

Multiply by $t^k e^{It}$ and take expectation:

$$\begin{aligned} & t^k e^{It} \mathbb{P}(Y_t(x) \geq ct + g(x), Y_t(y) \geq ct + g(y)) \\ &= C_{c,Q}^2 \mathbb{E}\left[\left(\frac{t}{\tau}\right)^k e^{I \log(1/\rho)} e^{-\lambda \cdot (a_x + a_y)} J_\tau(a_x) J_\tau(a_y)\right]. \end{aligned} \quad (17)$$

Now

$$-\lambda \cdot (a_x + a_y) = -2I \log \frac{1}{\rho} - \lambda \cdot (g(x) + g(y)) + \lambda \cdot (A_x + A_y),$$

since $\lambda \cdot c = c^\top Q^{-1} c = I$. Hence the prefactor inside the expectation equals

$$\left(\frac{t}{\tau}\right)^k \rho^I e^{-\lambda \cdot (g(x) + g(y))} e^{\lambda \cdot (A_x + A_y)} J_\tau(a_x) J_\tau(a_y).$$

We now let $t \rightarrow \infty$. Since ρ is fixed once (x, y) is fixed,

$$\tau = t - \log \frac{1}{\rho} \xrightarrow{t \rightarrow \infty} \infty, \quad \frac{t}{\tau} \rightarrow 1.$$

Also $J_\tau(a_x) J_\tau(a_y) \rightarrow 1$ almost surely.

To justify dominated convergence, note that for all large t ,

$$\left(\frac{t}{\tau}\right)^k \leq C_{x,y}, \quad 0 < J_\tau(a_x) J_\tau(a_y) \leq 1.$$

Therefore the integrand in (17) is bounded by

$$C_{x,y} \rho^I e^{-\lambda \cdot (g(x)+g(y))} e^{\lambda \cdot (A_x+A_y)}.$$

Because (A_x, A_y) is a fixed Gaussian pair, it remains to check that $\mathbb{E}[e^{\lambda \cdot (A_x+A_y)}] < \infty$, which is immediate.

By dominated convergence,

$$t^k e^{It} \mathbb{P}(\dots) \rightarrow C_{c,Q}^2 \rho^I e^{-\lambda \cdot (g(x)+g(y))} \mathbb{E}[e^{\lambda \cdot (A_x+A_y)}].$$

It remains to compute the Gaussian moment. Observe that

$$\lambda \cdot A_x = \lambda^\top T X^{(\text{co})}(x) = (T^\top \lambda) \cdot X^{(\text{co})}(x) = u_* \cdot X^{(\text{co})}(x),$$

and similarly

$$\lambda \cdot A_y = u_* \cdot X^{(\text{co})}(y).$$

Hence

$$\begin{aligned} \mathbb{E}[e^{\lambda \cdot (A_x+A_y)}] &= \exp\left(\frac{1}{2} \text{Var}(\lambda \cdot A_x + \lambda \cdot A_y)\right) \\ &= \exp\left(\frac{1}{2} \text{Var}(\lambda \cdot A_x) + \frac{1}{2} \text{Var}(\lambda \cdot A_y) + \text{Cov}(\lambda \cdot A_x, \lambda \cdot A_y)\right). \end{aligned}$$

Now

$$\text{Var}(\lambda \cdot A_x) = \text{Var}(u_* \cdot X^{(\text{co})}(x)) = I \log \frac{1}{\rho},$$

and the same for y . Therefore

$$\rho^I \exp\left(\frac{1}{2} \text{Var}(\lambda \cdot A_x) + \frac{1}{2} \text{Var}(\lambda \cdot A_y)\right) = \rho^I \exp\left(I \log \frac{1}{\rho}\right) = 1.$$

So the limit simplifies to

$$C_{c,Q}^2 e^{-\lambda \cdot (g(x)+g(y))} \exp\left(\text{Cov}(\lambda \cdot A_x, \lambda \cdot A_y)\right).$$

Finally,

$$\text{Cov}(\lambda \cdot A_x, \lambda \cdot A_y) = \text{Cov}(u_* \cdot X^{(\text{co})}(x), u_* \cdot X^{(\text{co})}(y)).$$

Since the fine fields at x and y are independent, the cross-covariance of the full field equals the cross-covariance of the coarse field:

$$\text{Cov}(u_* \cdot X^{(\text{co})}(x), u_* \cdot X^{(\text{co})}(y)) = K_\infty^{u_*}(x, y).$$

This proves (15). □

Lemma 4.7 (Gaussian shift identity). *Let (V, Z) be a jointly centered Gaussian vector, with V taking values in $\mathbb{R}^k$ and $Z \in \mathbb{R}$ having variance σ^2 . Then for every bounded measurable $F : \mathbb{R}^k \rightarrow \mathbb{R}$,*

$$\mathbb{E}[F(V) e^{Z - \frac{1}{2}\sigma^2}] = \mathbb{E}[F(V + \text{Cov}(V, Z))].$$

Proof. Define a probability measure $\tilde{\mathbb{P}}$ by

$$\frac{d\tilde{\mathbb{P}}}{d\mathbb{P}} = e^{Z - \frac{1}{2}\sigma^2}.$$

Since Z is centered Gaussian with variance σ^2 , we have

$$\mathbb{E}[e^{Z - \sigma^2/2}] = 1,$$

so $\tilde{\mathbb{P}}$ is indeed a probability measure.

To identify the law of V under $\tilde{\mathbb{P}}$, compute its characteristic function. For any $\theta \in \mathbb{R}^k$,

$$\tilde{\mathbb{E}}[e^{i\langle \theta, V \rangle}] = \mathbb{E}[e^{i\langle \theta, V \rangle} e^{Z - \frac{1}{2}\sigma^2}] = e^{-\frac{1}{2}\sigma^2} \mathbb{E}[e^{i\langle \theta, V \rangle + Z}].$$

Because (V, Z) is jointly *centered* Gaussian, the scalar random variable $i\langle \theta, V \rangle + Z$ is Gaussian. Therefore

$$\mathbb{E}[e^{i\langle \theta, V \rangle + Z}] = \exp\left(\frac{1}{2}\text{Var}(i\langle \theta, V \rangle + Z)\right).$$

Now

$$\text{Var}(i\langle \theta, V \rangle + Z) = -\text{Var}(\langle \theta, V \rangle) + 2i\text{Cov}(\langle \theta, V \rangle, Z) + \sigma^2,$$

so

$$\tilde{\mathbb{E}}[e^{i\langle \theta, V \rangle}] = \exp\left(-\frac{1}{2}\text{Var}(\langle \theta, V \rangle) + i\text{Cov}(\langle \theta, V \rangle, Z)\right).$$

On the other hand, the characteristic function of $V + \text{Cov}(V, Z)$ under $\mathbb{P}$ equals

$$\mathbb{E}[e^{i\langle \theta, V + \text{Cov}(V, Z) \rangle}] = \exp\left(i\langle \theta, \text{Cov}(V, Z) \rangle\right) \mathbb{E}[e^{i\langle \theta, V \rangle}] = \exp\left(i\text{Cov}(\langle \theta, V \rangle, Z) - \frac{1}{2}\text{Var}(\langle \theta, V \rangle)\right).$$

Thus V under $\tilde{\mathbb{P}}$ and $V + \text{Cov}(V, Z)$ under $\mathbb{P}$ have the same characteristic function, hence the same law. The claimed identity for bounded measurable F follows. $\square$

Proof of Theorem 4.4. Fix bounded continuous f . We prove

$$\mathbb{E}\left[\left(\int_D f \, d\nu_t - C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \mu_t^{u_*}(\mathrm{d}x)\right)^2\right] \xrightarrow[t \rightarrow \infty]{} 0. \quad (18)$$

Since Theorem 2.9 yields

$$\int_D f(x) e^{-\lambda \cdot g(x)} \mu_t^{u_*}(\mathrm{d}x) \xrightarrow[t \rightarrow \infty]{L^2} \int_D f(x) e^{-\lambda \cdot g(x)} \mu^{u_*}(\mathrm{d}x),$$

this implies the stated convergence.

We identify the three terms in the polarization identity.

Step 1: first moments. By Lemma 4.5 and dominated convergence,

$$\mathbb{E}\left[\int_D f \, d\nu_t\right] = \int_D f(x) t^{k/2} e^{It/2} \mathbb{P}(Y_t(x) \geq ct + g(x)) \, \mathrm{d}x \rightarrow C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \, \mathrm{d}x.$$

On the other hand, $\mathbb{E}[\mu_t^{u_*}(\mathrm{d}x)] = \mathrm{d}x$, so

$$\mathbb{E}\left[C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \mu_t^{u_*}(\mathrm{d}x)\right] = C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \, \mathrm{d}x.$$

Hence the first moments match.

Step 2: second moment of $\int f d\nu_t$. We write

$$\mathbb{E}\left[\left(\int_D f d\nu_t\right)^2\right] = \int_{D^2} f(x)f(y) t^k e^{It} \mathbb{P}(Y_t(x) \geq ct + g(x), Y_t(y) \geq ct + g(y)) dx dy.$$

For $x \neq y$, Lemma 4.6 gives pointwise convergence of the integrand to

$$f(x)f(y) C_{c,Q}^2 e^{-\lambda \cdot (g(x)+g(y))} \exp(K_\infty^{u*}(x, y)).$$

To justify dominated convergence, split D^2 into a small diagonal neighborhood and its complement. Let

$$\Delta_t := \{(x, y) \in D^2 : |x - y| \leq e^{-t/2}\}.$$

On Δ_t , the crude bound

$$\mathbb{P}(Y_t(x) \geq ct + g(x), Y_t(y) \geq ct + g(y)) \leq \mathbb{P}(Y_t(x) \geq ct + g(x))$$

and Lemma 4.5 imply that for all sufficiently large t ,

$$\sup_{x \in D} t^{k/2} e^{It/2} \mathbb{P}(Y_t(x) \geq ct + g(x)) \leq C,$$

because g is bounded and the asymptotic in Lemma 4.5 is uniform in x . Hence, after increasing C if necessary,

$$t^k e^{It} \mathbb{P}(\dots) \leq C t^{k/2} e^{It/2},$$

uniformly in x, y for all large t . Since $\text{Leb}(\Delta_t) \leq C_D e^{-dt/2}$, we obtain

$$\int_{\Delta_t} |f(x)f(y)| t^k e^{It} \mathbb{P}(\dots) dx dy \leq C \|f\|_\infty^2 t^{k/2} e^{(I-d)t/2} \xrightarrow{t \rightarrow \infty} 0,$$

because $I < d$.

On $D^2 \setminus \Delta_t$, we use a uniform off-diagonal bound. Fix $x \neq y$ and set $r = |x - y|$. Since $(x, y) \notin \Delta_t$, we have $r > e^{-t/2}$, hence

$$\rho = \min\{r/2, 1/2\} \geq e^{-t/2}/2.$$

Therefore

$$\log \frac{1}{\rho} \leq \frac{t}{2} + \log 2, \quad \tau = t - \log \frac{1}{\rho} \geq \frac{t}{2} - \log 2.$$

In particular, for all sufficiently large t ,

$$1 \leq \frac{t}{\tau} \leq 4,$$

uniformly over $(x, y) \in D^2 \setminus \Delta_t$. Thus from (17) and the estimate $0 < J_\tau \leq 1$, we get

$$t^k e^{It} \mathbb{P}(\dots) \leq C \rho^I e^{-\lambda \cdot (g(x)+g(y))} \mathbb{E}[e^{\lambda \cdot (A_x + A_y)}],$$

where C is now independent of x, y, t (for all sufficiently large t). As in the proof of Lemma 4.6,

$$\rho^I \mathbb{E}[e^{\lambda \cdot (A_x + A_y)}] = \exp(K_\infty^{u*}(x, y)).$$

By the logarithmic covariance estimate,

$$K_\infty^{u_*}(x, y) \leq I \log \frac{1}{|x - y|} + C_D \quad (x \neq y),$$

hence

$$\exp(K_\infty^{u_*}(x, y)) \leq C'_D |x - y|^{-I}.$$

Since $I < d$, the kernel $|x - y|^{-I}$ is integrable on D^2 . Therefore, if we write the integrand over all of D^2 and insert the indicator $\mathbf{1}_{\{|x - y| > e^{-t/2}\}}$, then the corresponding integrands converge pointwise for every $x \neq y$ and are dominated by the integrable kernel $C|f(x)f(y)|e^{-\lambda \cdot (g(x) + g(y))}|x - y|^{-I}$. Since the diagonal has Lebesgue measure 0, dominated convergence yields

$$\int_{D^2 \setminus \Delta_t} f(x)f(y) t^k e^{It} \mathbb{P}(Y_t(x) \geq ct + g(x), Y_t(y) \geq ct + g(y)) \, dx \, dy \rightarrow C_{c,Q}^2 \int_{D^2} f(x)f(y) e^{-\lambda \cdot (g(x) + g(y))} \exp(K_\infty^{u_*}(x, y)) \, dx \, dy.$$

Combining this with the vanishing of the near-diagonal contribution proves the desired second-moment limit.

Step 3: second moment of the GMC term. We must compute the second moment of the “scaled” GMC term appearing in (18), namely

$$C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \mu_t^{u_*}(\mathrm{d}x).$$

By the usual covariance formula for GMC,

$$\mathbb{E} \left[\left(\int_D f(x) e^{-\lambda \cdot g(x)} \mu_t^{u_*}(\mathrm{d}x) \right)^2 \right] = \int_{D^2} f(x)f(y) e^{-\lambda \cdot (g(x) + g(y))} \exp(\mathrm{Cov}(u_* \cdot X_t(x), u_* \cdot X_t(y))) \, dx \, dy.$$

Hence

$$\mathbb{E} \left[\left(C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \mu_t^{u_*}(\mathrm{d}x) \right)^2 \right] = C_{c,Q}^2 \int_{D^2} f(x)f(y) e^{-\lambda \cdot (g(x) + g(y))} \exp(\mathrm{Cov}(u_* \cdot X_t(x), u_* \cdot X_t(y))) \, dx \, dy.$$

For each fixed $x \neq y$,

$$\mathrm{Cov}(u_* \cdot X_t(x), u_* \cdot X_t(y)) \xrightarrow[t \rightarrow \infty]{} K_\infty^{u_*}(x, y).$$

Moreover, by the logarithmic covariance estimate, there exists $C_D < \infty$ such that for all $x \neq y$ and all $t \geq 0$,

$$\mathrm{Cov}(u_* \cdot X_t(x), u_* \cdot X_t(y)) \leq I \log \frac{1}{|x - y|} + C_D.$$

Hence

$$\exp(\mathrm{Cov}(u_* \cdot X_t(x), u_* \cdot X_t(y))) \leq C'_D |x - y|^{-I} \quad (x \neq y),$$

and the kernel on the right is integrable on D^2 because $I < d$. Therefore, if we regard the integrand as a function on the fixed space D^2 and ignore the diagonal (which has Lebesgue measure 0), dominated convergence gives

$$\mathbb{E} \left[\left(C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \mu_t^{u_*}(\mathrm{d}x) \right)^2 \right] \rightarrow C_{c,Q}^2 \int_{D^2} f(x)f(y) e^{-\lambda \cdot (g(x) + g(y))} \exp(K_\infty^{u_*}(x, y)) \, dx \, dy.$$

This is the same limit as in Step 2.

Step 4: cross term. We now compute

$$\mathbb{E} \left[\left(\int_D f \, d\nu_t \right) \left(C_{c,Q} \int_D f(y) e^{-\lambda \cdot g(y)} \mu_t^{u_*}(\mathrm{d}y) \right) \right].$$

Using Fubini and the density of $\mu_t^{u_*}$,

$$\begin{aligned} & \mathbb{E} \left[\left(\int_D f \, d\nu_t \right) \left(C_{c,Q} \int_D f(y) e^{-\lambda \cdot g(y)} \mu_t^{u_*}(\mathrm{d}y) \right) \right] \\ &= C_{c,Q} \int_{D^2} f(x) f(y) e^{-\lambda \cdot g(y)} t^{k/2} e^{It/2} \mathbb{E} \left[\mathbb{1}_{\{Y_t(x) \geq ct + g(x)\}} e^{u_* \cdot X_t(y) - It/2} \right] \mathrm{d}x \mathrm{d}y. \end{aligned}$$

Set

$$V_t(x) := Y_t(x) = TX_t(x) \in \mathbb{R}^k, \quad Z_t(y) := u_* \cdot X_t(y) \in \mathbb{R}.$$

Then $(V_t(x), Z_t(y))$ is jointly Gaussian with

$$\mathrm{Var}(Z_t(y)) = It, \quad \mathrm{Cov}(V_t(x), Z_t(y)) = T \mathbb{E}[X_t(x) X_t(y)^\top] u_*.$$

Since $\mathbb{E}[X_t(x) X_t(y)^\top] = \Sigma K_t(x, y)$, we get

$$\mathrm{Cov}(V_t(x), Z_t(y)) = T \Sigma u_* K_t(x, y) = c K_t(x, y).$$

Applying Lemma 4.7 with $F(v) = \mathbb{1}_{\{v \geq ct + g(x)\}}$, we obtain

$$\mathbb{E} \left[\mathbb{1}_{\{V_t(x) \geq ct + g(x)\}} e^{Z_t(y) - It/2} \right] = \mathbb{P}(V_t(x) + c K_t(x, y) \geq ct + g(x)).$$

Hence the cross term equals

$$C_{c,Q} \int_{D^2} f(x) f(y) e^{-\lambda \cdot g(y)} t^{k/2} e^{It/2} \mathbb{P}(V_t(x) \geq c(t - K_t(x, y)) + g(x)) \, \mathrm{d}x \mathrm{d}y. \quad (19)$$

For fixed $x \neq y$, define

$$a_t(x, y) := g(x) - c K_t(x, y).$$

Then $K_t(x, y) \rightarrow K_\infty(x, y) \in \mathbb{R}$, so $a_t(x, y) \rightarrow a_\infty(x, y) := g(x) - c K_\infty(x, y)$. Using Lemma 4.3,

$$t^{k/2} e^{It/2} \mathbb{P}(V_t(x) \geq ct + a_t(x, y)) = C_{c,Q} e^{-\lambda \cdot a_t(x, y)} J_t(a_t(x, y)),$$

where $0 < J_t(\cdot) \leq 1$ and

$$J_t(a) := \frac{\int_{[0, \infty)^k} e^{-\lambda \cdot z} \exp\left(-\frac{1}{2t}(a+z)^\top Q^{-1}(a+z)\right) \mathrm{d}z}{\int_{[0, \infty)^k} e^{-\lambda \cdot z} \mathrm{d}z}.$$

Since $a_t(x, y) \rightarrow a_\infty(x, y)$ and Q^{-1} is positive definite, dominated convergence gives

$$J_t(a_t(x, y)) \rightarrow 1 \quad (t \rightarrow \infty)$$

for each fixed $x \neq y$. Therefore the integrand in (19) converges pointwise on $D^2 \setminus \{x = y\}$ to

$$C_{c,Q}^2 f(x) f(y) e^{-\lambda \cdot (g(x) + g(y))} e^{IK_\infty(x, y)}.$$

It remains to justify dominated convergence.

Near-diagonal region. On $\Delta_t := \{(x, y) \in D^2 : |x - y| \leq e^{-t/2}\}$, the crude bound by 1 yields

$$t^{k/2} e^{It/2} \mathbb{P}(\dots) \leq t^{k/2} e^{It/2}.$$

Hence

$$\int_{\Delta_t} |f(x)f(y)| t^{k/2} e^{It/2} \mathbb{P}(\dots) \, dx \, dy \leq C \|f\|_\infty^2 t^{k/2} e^{(I-d)t/2} \xrightarrow[t \rightarrow \infty]{} 0,$$

since $I < d$.

Off-diagonal domination. On $D^2 \setminus \Delta_t$, we use the exact factorization again:

$$t^{k/2} e^{It/2} \mathbb{P}(V_t(x) \geq ct + a_t(x, y)) = C_{c,Q} e^{-\lambda \cdot a_t(x, y)} J_t(a_t(x, y)) \leq C_{c,Q} e^{-\lambda \cdot g(x)} e^{IK_t(x, y)},$$

because $J_t \leq 1$ and $\lambda \cdot c = I$. By the logarithmic covariance estimate, there exists $C_D < \infty$ such that for all $x \neq y$,

$$K_t(x, y) \leq \log \frac{1}{|x - y|} + C_D.$$

Since D is bounded, $1 \leq (\text{diam } D)^I |x - y|^{-I}$, so

$$e^{IK_t(x, y)} \leq C |x - y|^{-I} \quad \text{for all } x \neq y, \, t \geq 0.$$

Hence the integrand in (19) is dominated on $D^2 \setminus \Delta_t$ by

$$C |f(x)f(y)| e^{-\lambda \cdot (g(x) + g(y))} |x - y|^{-I},$$

which is integrable on D^2 because $I < d$.

Therefore, if we extend the integrand in (19) to all of D^2 by inserting the indicator $\mathbf{1}_{\{|x-y| > e^{-t/2}\}}$, then the resulting integrands converge pointwise for every $x \neq y$ and are dominated by the integrable kernel

$$C |f(x)f(y)| e^{-\lambda \cdot (g(x) + g(y))} |x - y|^{-I}.$$

Since the diagonal has Lebesgue measure 0, dominated convergence yields that the off-diagonal contribution converges to

$$C_{c,Q}^2 \int_{D^2} f(x)f(y) e^{-\lambda \cdot (g(x) + g(y))} \exp(K_\infty^{u*}(x, y)) \, dx \, dy.$$

This is the same limit as in Steps 2 and 3. By polarization we obtain (18). In particular, for every bounded continuous f ,

$$\nu_t(f) \xrightarrow[t \rightarrow \infty]{P} C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \mu^{u*}(dx).$$

To pass from convergence of test integrals to weak convergence of the random measures ν_t , we argue along arbitrary sequences. Let $t_n \rightarrow \infty$ be any deterministic sequence. Then the convergence above holds along (t_n) for every bounded continuous f , so Lemma 4.8 applied to the sequence $(\nu_{t_n})_{n \geq 1}$ yields

$$\nu_{t_n} \xrightarrow[n \rightarrow \infty]{P} C_{c,Q} e^{-\lambda \cdot g} \mu^{u*} \quad \text{weakly as finite measures.}$$

Applying Lemma 4.9 in the metric space of finite Borel measures on D equipped with the bounded-Lipschitz metric, we conclude that the full family $(\nu_t)_{t \geq 0}$ converges weakly in probability as $t \rightarrow \infty$, which is exactly (13). This completes the proof of Theorem 4.4. $\square$

Lemma 4.8 (From test functions to weak convergence in probability). *Let D be compact metric and let μ_n, μ be random finite Borel measures on D . Assume that for every bounded continuous $f : D \rightarrow \mathbb{R}$,*

$$\int_D f \, d\mu_n \xrightarrow[n \rightarrow \infty]{\mathbb{P}} \int_D f \, d\mu.$$

Then μ_n converges weakly to μ in probability.

Proof. Let

$$\text{BL}_1(D) := \{f : D \rightarrow \mathbb{R} : \|f\|_\infty \leq 1, \text{Lip}(f) \leq 1\},$$

and let d_{BL} be the bounded-Lipschitz metric

$$d_{\text{BL}}(\nu, \eta) := \sup_{f \in \text{BL}_1(D)} \left| \int f \, d\nu - \int f \, d\eta \right|.$$

On a compact metric space, d_{BL} metrizes the weak topology on finite Borel measures.

Fix $\varepsilon > 0$. By Arzelà–Ascoli, $\text{BL}_1(D)$ is compact in the sup norm, hence there exist $f_1, \dots, f_M \in \text{BL}_1(D)$ such that every $f \in \text{BL}_1(D)$ satisfies

$$\|f - f_j\|_\infty \leq \varepsilon \quad \text{for some } j \in \{1, \dots, M\}.$$

Therefore for any finite measures ν, η ,

$$d_{\text{BL}}(\nu, \eta) \leq \max_{1 \leq j \leq M} \left| \int f_j \, d\nu - \int f_j \, d\eta \right| + \varepsilon(\nu(D) + \eta(D)).$$

Applying this with $\nu = \mu_n$ and $\eta = \mu$, and noting that the constant function 1 is bounded continuous, we have

$$\mu_n(D) \xrightarrow{\mathbb{P}} \mu(D).$$

Hence the family $(\mu_n(D))_n$ is tight, and so is $\mu_n(D) + \mu(D)$.

Now let $\delta > 0$. We first verify that the family $(\mu_n(D) + \mu(D))_{n \geq 1}$ is tight. Since the constant function 1 is bounded and continuous, we have

$$\mu_n(D) \xrightarrow[n \rightarrow \infty]{\mathbb{P}} \mu(D).$$

Choose $R_0 < \infty$ such that

$$\mathbb{P}(\mu(D) > R_0) < \delta/4.$$

Then there exists N_0 such that for all $n \geq N_0$,

$$\mathbb{P}(|\mu_n(D) - \mu(D)| > R_0) < \delta/4.$$

Hence for every $n \geq N_0$,

$$\mathbb{P}(\mu_n(D) + \mu(D) > 3R_0) \leq \mathbb{P}(\mu(D) > R_0) + \mathbb{P}(|\mu_n(D) - \mu(D)| > R_0) < \delta/2.$$

For the finitely many indices $1 \leq n < N_0$, choose $R_1 < \infty$ such that

$$\max_{1 \leq n < N_0} \mathbb{P}(\mu_n(D) + \mu(D) > R_1) < \delta/2.$$

Set $R := \max\{3R_0, R_1\}$. Then

$$\sup_n \mathbb{P}(\mu_n(D) + \mu(D) > R) < \delta.$$

Therefore

$$\begin{aligned} \mathbb{P}(d_{\text{BL}}(\mu_n, \mu) > 2\varepsilon R + \delta) &\leq \mathbb{P}\left(\max_{1 \leq j \leq M} \left| \int f_j \, d\mu_n - \int f_j \, d\mu \right| > \delta\right) + \mathbb{P}(\mu_n(D) + \mu(D) > R) \\ &\leq \sum_{j=1}^M \mathbb{P}\left(\left| \int f_j \, d\mu_n - \int f_j \, d\mu \right| > \delta\right) + \delta. \end{aligned}$$

For each fixed j , the probability in the sum tends to 0 by assumption. Therefore

$$\limsup_{n \rightarrow \infty} \mathbb{P}(d_{\text{BL}}(\mu_n, \mu) > 2\varepsilon R + \delta) \leq \delta.$$

Since $\varepsilon, \delta > 0$ were arbitrary, this proves $d_{\text{BL}}(\mu_n, \mu) \rightarrow 0$ in probability, i.e. weak convergence in probability. $\square$

Lemma 4.9 (Sequential criterion for convergence in probability of a family). *Let (E, d) be a metric space, let $(X_t)_{t \geq 0}$ be E -valued random variables, and let X be an E -valued random variable. Assume that for every deterministic sequence $t_n \rightarrow \infty$ one has*

$$X_{t_n} \xrightarrow[n \rightarrow \infty]{P} X.$$

Then

$$X_t \xrightarrow[t \rightarrow \infty]{P} X.$$

Proof. Suppose the conclusion fails. Then there exist $\varepsilon_0 > 0$, $\delta_0 > 0$, and a deterministic sequence $t_n \rightarrow \infty$ such that

$$\mathbb{P}(d(X_{t_n}, X) > \varepsilon_0) \geq \delta_0 \quad \text{for all } n.$$

But this contradicts the assumed convergence in probability of X_{t_n} to X along every deterministic sequence. $\square$

5 The exact Hausdorff dimension in the L^2 -phase

We now complement the support statement with the matching upper bound in the L^2 -phase. Recall that, since $(t, x) \mapsto Y_t(x)$ is jointly continuous on every compact time interval, the pointwise limit set can be written using rational times. We therefore define

$$E(c) := \bigcap_{m \geq 1} \bigcup_{N \geq 1} \bigcap_{q \in \mathbb{Q} \cap [N, \infty)} \left\{ x \in D : \left| \frac{Y_q(x)}{q} - c \right| \leq \frac{1}{m} \right\}, \quad I := c^\top Q^{-1} c = \|u_*\|_\Sigma^2.$$

This is a Borel subset of D , and by continuity in t it coincides with the set of points x such that

$$\lim_{t \rightarrow \infty} \frac{Y_t(x)}{t} = c.$$

5.1 Lower bound via the rooted chaos

Let $\mathcal{F}_n := \sigma(X_s(\cdot) : 0 \leq s \leq n)$.

Lemma 5.1 (Conditional expectation identity for the limiting chaos). *Fix $n \in \mathbb{N}$. Let $h : D \times \Omega \rightarrow [0, \infty)$ be bounded and $\mathcal{F}_n \otimes \mathcal{B}(D)$ -measurable. Then*

$$\mathbb{E} \left[\int_D h(x, \omega) \mu^{u*}(\mathrm{d}x) \right] = \mathbb{E} \left[\int_D h(x, \omega) \mu_n^{u*}(\mathrm{d}x) \right].$$

Proof. We first prove the claim for simple tensors of the form

$$h(x, \omega) = \xi(\omega) \varphi(x),$$

where now ξ is bounded and $\mathcal{F}_n$ -measurable and φ is bounded and continuous (not necessarily nonnegative).

For such a φ , the process

$$\mu_t^{u*}(\varphi) := \int_D \varphi(x) \mu_t^{u*}(\mathrm{d}x)$$

is an L^2 -bounded martingale with respect to $(\mathcal{F}_t)_{t \geq 0}$. Indeed, by the same conditional-Fubini computation as in Lemma 2.8,

$$\mathbb{E}[\mu_t^{u*}(\varphi) \mid \mathcal{F}_n] = \mu_n^{u*}(\varphi) \quad (t \geq n).$$

Now let $m \rightarrow \infty$ along the integers. Since Theorem 2.9 gives

$$\mu_m^{u*}(\varphi) \xrightarrow[m \rightarrow \infty]{L^2, \text{ a.s.}} \mu^{u*}(\varphi),$$

we in particular have $\mu_m^{u*}(\varphi) \rightarrow \mu^{u*}(\varphi)$ in L^1 . Conditional expectation is continuous in L^1 , so

$$\mathbb{E}[\mu^{u*}(\varphi) \mid \mathcal{F}_n] = \lim_{m \rightarrow \infty} \mathbb{E}[\mu_m^{u*}(\varphi) \mid \mathcal{F}_n] = \mu_n^{u*}(\varphi) \quad \text{a.s.}$$

Therefore, for every bounded $\mathcal{F}_n$ -measurable ξ ,

$$\mathbb{E}[\xi \mu^{u*}(\varphi)] = \mathbb{E}[\xi \mathbb{E}[\mu^{u*}(\varphi) \mid \mathcal{F}_n]] = \mathbb{E}[\xi \mu_n^{u*}(\varphi)].$$

Thus the desired identity holds for all such simple tensors.

Now let $\mathcal{C}_n$ be the collection of bounded Borel functions $\varphi : D \rightarrow \mathbb{R}$ such that

$$\mathbb{E}[\xi \mu^{u*}(\varphi)] = \mathbb{E}[\xi \mu_n^{u*}(\varphi)]$$

for every bounded $\mathcal{F}_n$ -measurable ξ . By the argument above, $\mathcal{C}_n$ contains every bounded continuous φ . Moreover, $\mathcal{C}_n$ is a vector space. It is also closed under bounded monotone convergence of nonnegative functions: if $0 \leq \varphi_m \uparrow \varphi$ bounded and each $\varphi_m \in \mathcal{C}_n$, then for every bounded nonnegative $\mathcal{F}_n$ -measurable ξ , the monotone convergence theorem gives

$$\mathbb{E}[\xi \mu^{u*}(\varphi_m)] \uparrow \mathbb{E}[\xi \mu^{u*}(\varphi)], \quad \mathbb{E}[\xi \mu_n^{u*}(\varphi_m)] \uparrow \mathbb{E}[\xi \mu_n^{u*}(\varphi)].$$

By linearity in ξ , the same identity then holds for all bounded signed $\mathcal{F}_n$ -measurable ξ . Since bounded continuous functions form an algebra generating $\mathcal{B}(D)$ on the compact metric space

D , the functional monotone class theorem implies that $\mathcal{C}_n$ contains all bounded Borel functions $\varphi: D \rightarrow \mathbb{R}$.

Finally, let $\mathcal{H}_n$ be the collection of bounded $\mathcal{F}_n \otimes \mathcal{B}(D)$ -measurable functions $h: D \times \Omega \rightarrow \mathbb{R}$ such that

$$\mathbb{E} \left[\int_D h(x, \omega) \mu^{u_*}(dx) \right] = \mathbb{E} \left[\int_D h(x, \omega) \mu_n^{u_*}(dx) \right].$$

By what we just proved, $\mathcal{H}_n$ contains all bounded simple tensors $\xi(\omega)\varphi(x)$ with ξ bounded $\mathcal{F}_n$ -measurable and φ bounded Borel. The class $\mathcal{H}_n$ is a vector space and is closed under bounded monotone pointwise convergence of nonnegative functions by the monotone convergence theorem. Hence, by the functional monotone class theorem on the product space $D \times \Omega$, $\mathcal{H}_n$ contains every bounded $\mathcal{F}_n \otimes \mathcal{B}(D)$ -measurable function. In particular it contains every bounded nonnegative such function, proving the lemma. $\square$

Lemma 5.2 (Rooted law at a point). *Fix $x \in D$ and $n \in \mathbb{N}$. Define a probability measure $\mathbb{Q}_{n,x}$ on Ω by*

$$\frac{d\mathbb{Q}_{n,x}}{d\mathbb{P}} = \exp \left(u_* \cdot X_n(x) - \frac{1}{2} I n \right), \quad I = \|u_*\|_\Sigma^2.$$

Then for every finite collection $0 \leq s_1 < \dots < s_\ell \leq n$, the vector

$$(Y_{s_1}(x), \dots, Y_{s_\ell}(x))$$

under $\mathbb{Q}_{n,x}$ has the same law as

$$(Y_{s_1}(x) + c s_1, \dots, Y_{s_\ell}(x) + c s_\ell)$$

under $\mathbb{P}$. In particular, under $\mathbb{Q}_{n,x}$,

$$Y_n(x) - c n \sim \mathcal{N}(0, nQ).$$

Proof. Fix $0 \leq s_1 < \dots < s_\ell \leq n$. By Lemma 4.7, it is enough to compute the covariance shift:

$$\text{Cov}(Y_{s_j}(x), u_* \cdot X_n(x)) = T \mathbb{E}[X_{s_j}(x) X_n(x)^\top] u_*.$$

Since $x = x$ and $s_j \leq n$, Lemma 2.4 gives

$$\mathbb{E}[X_{s_j}(x) X_n(x)^\top] = \Sigma K_{s_j}(x, x) = \Sigma s_j.$$

Therefore

$$\text{Cov}(Y_{s_j}(x), u_* \cdot X_n(x)) = T \Sigma u_* s_j = c s_j.$$

Applying Lemma 4.7 to the finite-dimensional Gaussian vector

$$(Y_{s_1}(x), \dots, Y_{s_\ell}(x), u_* \cdot X_n(x))$$

yields the claimed shift of law. $\square$

Proposition 5.3 (Support and conditional lower Hausdorff bound in the L^2 -phase). *Assume the hypotheses of Theorem 4.4. Then almost surely*

$$\mu^{u_*}(D \setminus E(c)) = 0.$$

In particular, on the event $\{\mu^{u_}(D) > 0\}$,*

$$\dim_H E(c) \geq d - \frac{1}{2} I.$$

Proof. Fix $\eta > 0$. For each $n \in \mathbb{N}$ and each $x \in D$, define the events

$$\mathcal{B}_n^\eta(x) := \left\{ \omega : \left| \frac{Y_n(x, \omega)}{n} - c \right| > \eta \right\},$$

and

$$\mathcal{C}_n^\eta(x) := \left\{ \omega : \sup_{s \in [n, n+1]} |(Y_s(x) - cs) - (Y_n(x) - cn)| > \eta n \right\}.$$

For each realization ω , define the corresponding random subsets of D by

$$B_n^\eta(\omega) := \{x \in D : \omega \in \mathcal{B}_n^\eta(x)\}, \quad C_n^\eta(\omega) := \{x \in D : \omega \in \mathcal{C}_n^\eta(x)\}.$$

Step 1: summability of $\mathbb{E}[\mu^{u}(B_n^\eta)]$.* Since $Y_n(\cdot)$ is jointly measurable on $D \times \Omega$, the map $(x, \omega) \mapsto \mathbf{1}_{\mathcal{B}_n^\eta(x)}(\omega)$ is bounded and $\mathcal{F}_n \otimes \mathcal{B}(D)$ -measurable. Hence Lemma 5.1 gives

$$\mathbb{E}[\mu^{u*}(B_n^\eta)] = \mathbb{E} \left[\int_D \mathbf{1}_{\mathcal{B}_n^\eta(x)} \mu_n^{u*}(dx) \right] = \int_D \mathbb{Q}_{n,x}(\mathcal{B}_n^\eta(x)) dx.$$

By Lemma 5.2, under $\mathbb{Q}_{n,x}$ we have

$$Y_n(x) - cn \sim \mathcal{N}(0, nQ),$$

independently of x . Therefore

$$\mathbb{Q}_{n,x}(\mathcal{B}_n^\eta(x)) = \mathbb{P}(|Z_n| > \eta n), \quad Z_n \sim \mathcal{N}(0, nQ).$$

Since Q is positive definite, there exist $C, c > 0$ such that

$$\mathbb{P}(|Z_n| > \eta n) \leq C e^{-c\eta^2 n}.$$

Hence

$$\mathbb{E}[\mu^{u*}(B_n^\eta)] \leq C_D e^{-c\eta^2 n},$$

and in particular

$$\sum_{n \geq 1} \mathbb{E}[\mu^{u*}(B_n^\eta)] < \infty.$$

Step 2: summability of $\mathbb{E}[\mu^{u}(C_n^\eta)]$.* By Lemma 2.5, we work with a jointly measurable version of $(s, x) \mapsto Y_s(x)$ that is continuous in s for every fixed x . Hence

$$\sup_{s \in [n, n+1]} |(Y_s(x) - cs) - (Y_n(x) - cn)| = \sup_{s \in ([n, n+1] \cap \mathbb{Q})} |(Y_s(x) - cs) - (Y_n(x) - cn)|,$$

so the map $(x, \omega) \mapsto \mathbf{1}_{\mathcal{C}_n^\eta(x)}(\omega)$ is bounded and $\mathcal{F}_{n+1} \otimes \mathcal{B}(D)$ -measurable. Therefore Lemma 5.1 yields

$$\mathbb{E}[\mu^{u*}(C_n^\eta)] = \mathbb{E} \left[\int_D \mathbf{1}_{\mathcal{C}_n^\eta(x)} \mu_{n+1}^{u*}(dx) \right] = \int_D \mathbb{Q}_{n+1,x}(\mathcal{C}_n^\eta(x)) dx.$$

Fix $x \in D$. Under $\mathbb{Q}_{n+1,x}$, Lemma 5.2 implies that the finite-dimensional laws of

$$M_s(x) := Y_s(x) - cs, \quad 0 \leq s \leq n+1,$$

coincide with those of the centered Gaussian process $Y_s(x)$ under $\mathbb{P}$. We now justify the passage to path-space laws. By Lemma 2.5, we have fixed a version of $s \mapsto Y_s(x)$ that is continuous

on $[n, n+1]$ almost surely under $\mathbb{P}$. Since $\mathbb{Q}_{n+1,x}$ is absolutely continuous with respect to $\mathbb{P}$, the same version is also continuous on $[n, n+1]$ almost surely under $\mathbb{Q}_{n+1,x}$. Therefore both path-valued random variables

$$s \mapsto M_s(x) \quad \text{under } \mathbb{Q}_{n+1,x}, \quad s \mapsto Y_s(x) \quad \text{under } \mathbb{P},$$

are $C([n, n+1], \mathbb{R}^k)$ -valued. Since the Borel σ -field of $C([n, n+1], \mathbb{R}^k)$ is generated by the evaluation maps at rational times, equality of finite-dimensional distributions on the countable dense set $([n, n+1] \cap \mathbb{Q})$ implies equality of the induced laws on $C([n, n+1], \mathbb{R}^k)$. In particular, for $r, s \in [n, n+1]$,

$$M_s(x) - M_r(x) \stackrel{d}{=} Y_s(x) - Y_r(x) = T(X_s(x) - X_r(x)).$$

Write $M_s(x) = (M_s^{(1)}(x), \dots, M_s^{(k)}(x))$. For each coordinate j ,

$$M_s^{(j)}(x) - M_r^{(j)}(x) \stackrel{d}{=} Y_s^{(j)}(x) - Y_r^{(j)}(x),$$

which is a centered real Gaussian increment with variance bounded by $C_Q |s-r|$, uniformly in j, x , and $r, s \in [n, n+1]$. Thus the canonical metric of each scalar process $(M_s^{(j)}(x) - M_n^{(j)}(x))_{s \in [n, n+1]}$ is dominated by $C_Q^{1/2} |s-r|^{1/2}$.

Since the parameter space $[n, n+1]$ has unit length, Dudley's entropy integral is finite and bounded uniformly in n, x, j . Therefore

$$m_Q := \sup_{n \geq 1} \sup_{x \in D} \sup_{1 \leq j \leq k} \mathbb{E}_{\mathbb{Q}_{n+1,x}} \left[\sup_{s \in [n, n+1]} |M_s^{(j)}(x) - M_n^{(j)}(x)| \right] < \infty.$$

By Borell–TIS for each scalar coordinate process, there exist $\sigma_Q^2 < \infty$ and constants $C, c > 0$ such that for all $u > 0$,

$$\mathbb{Q}_{n+1,x} \left(\sup_{s \in [n, n+1]} |M_s^{(j)}(x) - M_n^{(j)}(x)| \geq m_Q + u \right) \leq C \exp \left(- \frac{u^2}{2\sigma_Q^2} \right),$$

uniformly in n, x, j . Since

$$\sup_{s \in [n, n+1]} |M_s(x) - M_n(x)| \leq \sum_{j=1}^k \sup_{s \in [n, n+1]} |M_s^{(j)}(x) - M_n^{(j)}(x)|,$$

the event $\mathcal{C}_n^\eta(x)$ implies that for some j ,

$$\sup_{s \in [n, n+1]} |M_s^{(j)}(x) - M_n^{(j)}(x)| > \frac{\eta n}{k}.$$

For n large enough so that $\eta n/k \geq 2m_Q$, we may set $u = \eta n/k - m_Q$ and obtain, by a union bound over j ,

$$\mathbb{Q}_{n+1,x}(\mathcal{C}_n^\eta(x)) \leq C e^{-c\eta^2 n^2},$$

for new constants $C, c > 0$ depending only on Q and k . For the finitely many smaller n , the trivial bound

$$\mathbb{Q}_{n+1,x}(\mathcal{C}_n^\eta(x)) \leq 1$$

holds. Hence, after enlarging C if necessary, we may assume that

$$\mathbb{Q}_{n+1,x}(C_n^\eta(x)) \leq C e^{-c\eta^2 n^2} \quad \text{for all } n \geq 1.$$

Therefore

$$\mathbb{E}[\mu^{u*}(C_n^\eta)] \leq C'_D e^{-c\eta^2 n^2},$$

and hence

$$\sum_{n \geq 1} \mathbb{E}[\mu^{u*}(C_n^\eta)] < \infty.$$

Step 3: Borel–Cantelli for the random measure μ^{u} .* Since

$$\sum_{n \geq 1} \mathbb{E}[\mu^{u*}(B_n^\eta)] < \infty, \quad \sum_{n \geq 1} \mathbb{E}[\mu^{u*}(C_n^\eta)] < \infty,$$

we claim that

$$\mu^{u*}\left(\limsup_{n \rightarrow \infty} B_n^\eta\right) = 0, \quad \mu^{u*}\left(\limsup_{n \rightarrow \infty} C_n^\eta\right) = 0 \quad \text{a.s.}$$

Indeed, for every N ,

$$\limsup_{n \rightarrow \infty} B_n^\eta \subset \bigcup_{n \geq N} B_n^\eta,$$

hence by subadditivity of the random measure μ^{u*} ,

$$\mu^{u*}\left(\limsup_{n \rightarrow \infty} B_n^\eta\right) \leq \mu^{u*}\left(\bigcup_{n \geq N} B_n^\eta\right) \leq \sum_{n \geq N} \mu^{u*}(B_n^\eta).$$

Since the right-hand side decreases as $N \rightarrow \infty$, taking the limit and then expectations gives

$$\mathbb{E}\left[\mu^{u*}\left(\limsup_{n \rightarrow \infty} B_n^\eta\right)\right] \leq \liminf_{N \rightarrow \infty} \sum_{n \geq N} \mathbb{E}[\mu^{u*}(B_n^\eta)] = 0.$$

Because μ^{u*} is nonnegative, it follows that

$$\mu^{u*}\left(\limsup_{n \rightarrow \infty} B_n^\eta\right) = 0 \quad \text{a.s.}$$

The same argument applies to C_n^η , yielding

$$\mu^{u*}\left(\limsup_{n \rightarrow \infty} C_n^\eta\right) = 0 \quad \text{a.s.}$$

Step 4: identification of the limit. Fix x outside $\limsup B_n^\eta \cup \limsup C_n^\eta$. Then there exists $N(x)$ such that for all $n \geq N(x)$,

$$\left| \frac{Y_n(x)}{n} - c \right| \leq \eta,$$

and

$$\sup_{s \in [n, n+1]} |(Y_s(x) - cs) - (Y_n(x) - cn)| \leq \eta n.$$

Hence for all $s \in [n, n+1]$ with $n \geq N(x)$,

$$|Y_s(x) - cs| \leq |Y_n(x) - cn| + \eta n \leq 2\eta n.$$

Dividing by $s \geq n$ gives

$$\left| \frac{Y_s(x)}{s} - c \right| = \frac{|Y_s(x) - cs|}{s} \leq 2\eta.$$

Therefore

$$\limsup_{s \rightarrow \infty} \left| \frac{Y_s(x)}{s} - c \right| \leq 2\eta.$$

Now fix a decreasing sequence $\eta_m \downarrow 0$, for instance $\eta_m = 1/m$, and intersect the corresponding full-probability events obtained above for each m . On this intersection, for every m there exists a μ^{u*} -null set $N_m \subset D$ such that for all $x \in D \setminus N_m$,

$$\limsup_{t \rightarrow \infty} \left| \frac{Y_t(x)}{t} - c \right| \leq 2\eta_m.$$

Since $\bigcup_{m \geq 1} N_m$ is still μ^{u*} -null, it follows that for μ^{u*} -a.e. x , the above bound holds simultaneously for every m . Letting $m \rightarrow \infty$ then yields

$$\lim_{t \rightarrow \infty} \frac{Y_t(x)}{t} = c.$$

Thus $x \in E(c)$ for μ^{u*} -a.e. x , and therefore

$$\mu^{u*}(D \setminus E(c)) = 0 \quad \text{almost surely.}$$

Step 5: Hausdorff lower bound. By exact-dimensionality of subcritical GMC, μ^{u*} has almost sure local dimension $d - \frac{1}{2}I$; see [16, Theorem 3.1 and Proposition 3.4]. Since $\mu^{u*}(D \setminus E(c)) = 0$, on the event $\{\mu^{u*}(D) > 0\}$ we have $\mu^{u*}(E(c)) = \mu^{u*}(D) > 0$. Therefore Frostman's lemma implies

$$\dim_H E(c) \geq d - \frac{1}{2}I \quad \text{on the event } \{\mu^{u*}(D) > 0\}.$$

This completes the proof. $\square$

5.2 A uniform oscillation estimate on a regular spatial grid

For $n \in \mathbb{N}$, set $r_n := e^{-n}$, and let $\mathcal{Q}_n$ be the regular grid partition of $D = [0, 1]^d$ into half-open axis-parallel cubes of mesh r_n , with the usual smaller boundary cubes along the faces of D . In particular, every $Q \in \mathcal{Q}_n$ has side length at most r_n , and

$$|\mathcal{Q}_n| \asymp r_n^{-d} = e^{dn}.$$

For $Q \in \mathcal{Q}_n$, define the oscillation of Y_n on Q by

$$\text{Osc}_n(Q) := \sup_{x, y \in Q} |Y_n(x) - Y_n(y)|,$$

where $|\cdot|$ denotes the Euclidean norm in $\mathbb{R}^k$.

Lemma 5.4 (Uniform subgaussian oscillation on scale e^{-n}). *There exist constants $C, c \in (0, \infty)$ such that for every $n \geq 1$, every $Q \in \mathcal{Q}_n$, and every $u > 0$,*

$$\mathbb{P}(\text{Osc}_n(Q) > u) \leq C e^{-cu^2}.$$

In particular, for every $\eta > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}(\exists Q \in \mathcal{Q}_n : \text{Osc}_n(Q) > \eta n) < \infty.$$

Proof. Fix $n \geq 1$ and $Q \in \mathcal{Q}_n$. Write $r_n = e^{-n}$ and let x_Q be the lower-left corner of Q . Even when Q is a smaller boundary cube, the point $x_Q + r_n z$ may lie outside D ; this is harmless because the field X_n , hence also $Y_n = TX_n$, has been defined on all of $\mathbb{R}^d$ by the same white-noise integral (1). For $z \in [0, 1]^d$, set

$$\tilde{Y}_n(z) := Y_n(x_Q + r_n z).$$

Since $Q \subset x_Q + [0, r_n]^d$, we have the deterministic bound

$$\text{Osc}_n(Q) \leq \sup_{z, z' \in [0, 1]^d} |\tilde{Y}_n(z) - \tilde{Y}_n(z')|.$$

Thus it suffices to control the oscillation of $\tilde{Y}_n$ on the whole reference cube $[0, 1]^d$.

We first bound the canonical metric of $\tilde{Y}_n$. Let $z, z' \in [0, 1]^d$, and set $h = z - z'$, so $|h| \leq \sqrt{d}$. By stationarity of the kernel and the definition $Y_n = TX_n$,

$$\mathbb{E}|\tilde{Y}_n(z) - \tilde{Y}_n(z')|^2 = 2 \text{Tr}(T \Sigma T^\top) \int_{e^{-n}}^1 \left(k(0) - k\left(\frac{r_n h}{r}\right) \right) \frac{dr}{r}.$$

Changing variables $r = r_n s$ gives

$$\mathbb{E}|\tilde{Y}_n(z) - \tilde{Y}_n(z')|^2 = 2 \text{Tr}(T \Sigma T^\top) \int_1^{e^n} \left(k(0) - k\left(\frac{h}{s}\right) \right) \frac{ds}{s}.$$

Since $k \in C_c^\infty(\mathbb{R}^d)$ is even and supported in $B(0, 1/2)$, there exists $C_0 < \infty$ such that

$$0 \leq k(0) - k(w) \leq C_0 |w|^2 \quad \text{for all } |w| \leq \frac{1}{4}.$$

We distinguish two cases.

Case 1: $|h| \leq \frac{1}{4}$. Then $|h|/s \leq \frac{1}{4}$ for all $s \geq 1$, hence

$$k(0) - k(h/s) \leq C_0 |h|^2 s^{-2}.$$

Therefore

$$\int_1^{e^n} \left(k(0) - k(h/s) \right) \frac{ds}{s} \leq C_0 |h|^2 \int_1^{e^n} s^{-3} ds \leq C_1 |h|^2.$$

Case 2: $|h| > \frac{1}{4}$. We split according to whether $4|h| \leq e^n$.

If $4|h| \geq e^n$, then on the whole interval $[1, e^n]$ we simply use boundedness of k :

$$0 \leq k(0) - k(h/s) \leq 2\|k\|_\infty,$$

so

$$\int_1^{e^n} \left(k(0) - k(h/s) \right) \frac{ds}{s} \leq 2\|k\|_\infty n.$$

Since $e^n \leq 4|h|$ in this case, we have $n \leq \log(4|h|)$. As $|h| \in [1/4, \sqrt{d}]$, the quantity $\log(4|h|)$ is bounded above by a constant depending only on d . Because $|h|^2 \geq 1/16$, it follows that

$$2\|k\|_\infty n \leq C_2 \leq C_3 |h|^2.$$

If $4|h| < e^n$, then we may split at $4|h|$:

$$\int_1^{e^n} \left(k(0) - k(h/s) \right) \frac{ds}{s} = \int_1^{4|h|} \left(k(0) - k(h/s) \right) \frac{ds}{s} + \int_{4|h|}^{e^n} \left(k(0) - k(h/s) \right) \frac{ds}{s}.$$

On $[1, 4|h|]$, boundedness of k gives

$$\int_1^{4|h|} \left(k(0) - k(h/s) \right) \frac{ds}{s} \leq 2\|k\|_\infty \log(4|h|) \leq C_4.$$

Again $|h| \geq \frac{1}{4}$, so $C_4 \leq C_5|h|^2$. On $[4|h|, e^n]$ we have $|h|/s \leq \frac{1}{4}$, hence

$$\int_{4|h|}^{e^n} \left(k(0) - k(h/s) \right) \frac{ds}{s} \leq C_0|h|^2 \int_{4|h|}^\infty s^{-3} ds \leq C_6|h|^2.$$

Combining the two pieces gives

$$\int_1^{e^n} \left(k(0) - k(h/s) \right) \frac{ds}{s} \leq C_7|h|^2.$$

In both cases,

$$\mathbb{E}|\tilde{Y}_n(z) - \tilde{Y}_n(z')|^2 \leq C_6|z - z'|^2 \quad \text{for all } z, z' \in [0, 1]^d,$$

with C_6 independent of n and Q . Thus each coordinate process of $\tilde{Y}_n$ has canonical metric uniformly dominated by a Euclidean metric on $[0, 1]^d$.

For $j = 1, \dots, k$, let $\tilde{Y}_n^{(j)}$ denote the j -th coordinate of $\tilde{Y}_n$, and define the scalar oscillation

$$\text{Osc}_n^{(j)}(Q) := \sup_{z, z' \in [0, 1]^d} |\tilde{Y}_n^{(j)}(z) - \tilde{Y}_n^{(j)}(z')|.$$

By the bound $\mathbb{E}|\tilde{Y}_n(z) - \tilde{Y}_n(z')|^2 \leq C_6|z - z'|^2$, each scalar coordinate satisfies

$$\mathbb{E}|\tilde{Y}_n^{(j)}(z) - \tilde{Y}_n^{(j)}(z')|^2 \leq C_6|z - z'|^2.$$

Hence Dudley's entropy bound implies

$$\sup_{n, Q, j} \mathbb{E}[\text{Osc}_n^{(j)}(Q)] < \infty.$$

Also, for every fixed $z \in [0, 1]^d$,

$$\text{Var}(\tilde{Y}_n^{(j)}(z) - \tilde{Y}_n^{(j)}(0)) \leq C_6|z|^2 \leq C_6d,$$

uniformly in n, Q, j . Therefore Borell–TIS applied to each scalar coordinate process yields constants $c, C > 0$, independent of n, Q, j , such that

$$\mathbb{P}(\text{Osc}_n^{(j)}(Q) > u) \leq Ce^{-cu^2}, \quad u > 0.$$

Now

$$\text{Osc}_n(Q) \leq \sum_{j=1}^k \text{Osc}_n^{(j)}(Q),$$

so if $\text{Osc}_n(Q) > u$, then for some j one has $\text{Osc}_n^{(j)}(Q) > u/k$. By a union bound,

$$\mathbb{P}(\text{Osc}_n(Q) > u) \leq kCe^{-c(u/k)^2}.$$

Absorbing k into the constants gives

$$\mathbb{P}(\text{Osc}_n(Q) > u) \leq Ce^{-cu^2}, \quad u > 0.$$

Finally, since $|\mathcal{Q}_n| \asymp e^{dn}$, another union bound gives

$$\mathbb{P}(\exists Q \in \mathcal{Q}_n : \text{Osc}_n(Q) > \eta n) \leq e^{dn} Ce^{-c\eta^2 n^2},$$

and the right-hand side is summable in n . □

5.3 A Gaussian vector thick-point upper bound

For $\eta > 0$, define the η -approximate thick-point limsup set

$$E_\eta := \left\{ x \in D : \left| \frac{1}{n} Y_n(x) - c \right| \leq \eta \text{ for infinitely many } n \right\}.$$

Clearly,

$$E(c) \subset E_\eta \quad \text{for every } \eta > 0. \quad (20)$$

Proposition 5.5 (Upper Hausdorff bound for vector thick points). *Let*

$$\Psi(\eta) := \inf \left\{ \frac{1}{2} a^\top Q^{-1} a : a \in \mathbb{R}^k, |a - c| \leq 2\eta \right\},$$

where $|\cdot|$ is the Euclidean norm on $\mathbb{R}^k$. Then:

- if $\Psi(\eta) > d$, the set E_η is empty almost surely;
- in all cases,

$$\dim_H(E_\eta) \leq \max\{d - \Psi(\eta), 0\} \quad \text{almost surely.}$$

Consequently,

$$\dim_H E(c) \leq d - \frac{1}{2} c^\top Q^{-1} c = d - \frac{1}{2} I \quad \text{almost surely.}$$

Proof. Fix $\eta > 0$. By Lemma 5.4 and Borel–Cantelli, there exists an almost sure event Ω_η such that for every $\omega \in \Omega_\eta$, for all sufficiently large n and all $Q \in \mathcal{Q}_n$,

$$\text{Osc}_n(Q; \omega) \leq \eta n.$$

Fix $\omega \in \Omega_\eta$ and let $x \in E_\eta(\omega)$. By definition, there exist infinitely many n such that

$$|Y_n(x, \omega) - cn| \leq \eta n.$$

For each such n , let $Q_n(x) \in \mathcal{Q}_n$ be the unique grid cube containing x , and let z_Q denote the center of a cube $Q \in \mathcal{Q}_n$. In particular, $z_{Q_n(x)}$ is the center of $Q_n(x)$. Since $\text{Osc}_n(Q_n(x); \omega) \leq \eta n$, we get

$$|Y_n(z_{Q_n(x)}, \omega) - cn| \leq 2\eta n.$$

Hence, for every $\omega \in \Omega_\eta$,

$$E_\eta(\omega) \subset \limsup_{n \rightarrow \infty} \bigcup_{Q \in \mathcal{G}_n^\eta(\omega)} Q,$$

where

$$\mathcal{G}_n^\eta(\omega) := \left\{ Q \in \mathcal{Q}_n : |Y_n(z_Q, \omega) - cn| \leq 2\eta n \right\}, \quad N_n^\eta(\omega) := |\mathcal{G}_n^\eta(\omega)|.$$

We first estimate $\mathbb{E}[N_n^\eta]$. Since every $Y_n(z_Q)$ has law $\mathcal{N}(0, nQ)$,

$$\mathbb{E}[N_n^\eta] = |\mathcal{Q}_n| \mathbb{P}(|Z_n - cn| \leq 2\eta n), \quad Z_n \sim \mathcal{N}(0, nQ).$$

The Gaussian density of Z_n is

$$p_n(z) = \frac{1}{(2\pi n)^{k/2} \sqrt{\det Q}} \exp\left(-\frac{1}{2n} z^\top Q^{-1} z\right).$$

Hence

$$\begin{aligned}\mathbb{P}(|Z_n - cn| \leq 2\eta n) &= \int_{|z-cn| \leq 2\eta n} p_n(z) dz \\ &\leq \frac{|B(0, 2\eta n)|}{(2\pi n)^{k/2} \sqrt{\det Q}} \sup_{|z-cn| \leq 2\eta n} \exp\left(-\frac{1}{2n} z^\top Q^{-1} z\right).\end{aligned}$$

Write $z = na$. Then $|a - c| \leq 2\eta$, and the supremum becomes

$$\exp\left(-n \inf_{|a-c| \leq 2\eta} \frac{1}{2} a^\top Q^{-1} a\right) = e^{-n\Psi(\eta)}.$$

Also $|B(0, 2\eta n)| \leq C_\eta n^k$, so

$$\mathbb{P}(|Z_n - cn| \leq 2\eta n) \leq C'_\eta n^{k/2} e^{-n\Psi(\eta)}.$$

Since $|\mathcal{Q}_n| \asymp e^{dn}$, we obtain

$$\mathbb{E}[N_n^\eta] \leq C''_\eta n^{k/2} e^{n(d-\Psi(\eta))}.$$

We now distinguish two cases.

Case 1: $\Psi(\eta) > d$. Then $\sum_n \mathbb{E}[N_n^\eta] < \infty$. By Markov and Borel–Cantelli, there exists an almost sure event Ω''_η such that for every $\omega \in \Omega''_\eta$, $N_n^\eta(\omega) = 0$ for all sufficiently large n . On $\Omega_\eta \cap \Omega''_\eta$, the covering relation above implies $E_\eta(\omega) = \emptyset$. Thus E_η is empty almost surely.

Case 2: $\Psi(\eta) \leq d$. For each integer $m \geq 1$, choose a rational number

$$s_m > d - \Psi(\eta), \quad s_m \downarrow d - \Psi(\eta).$$

We claim that for each m ,

$$\mathcal{H}^{s_m}(E_\eta(\omega)) = 0 \quad \text{for almost every } \omega.$$

Fix m . It suffices to show that

$$\sum_{n=1}^{\infty} \sum_{Q \in \mathcal{Q}_n^\eta(\omega)} (\text{diam } Q)^{s_m} < \infty \quad \text{for almost every } \omega.$$

Since $\text{diam } Q \leq \sqrt{d} e^{-n}$ for $Q \in \mathcal{Q}_n$, it is enough to prove

$$\sum_{n=1}^{\infty} e^{-ns_m} N_n^\eta(\omega) < \infty \quad \text{for almost every } \omega.$$

By nonnegativity, it is enough to show

$$\sum_{n=1}^{\infty} \mathbb{E}[e^{-ns_m} N_n^\eta] < \infty.$$

Using the bound on $\mathbb{E}[N_n^\eta]$,

$$\sum_{n=1}^{\infty} \mathbb{E}[e^{-ns_m} N_n^\eta] \leq C''_\eta \sum_{n=1}^{\infty} n^{k/2} e^{-n(s_m - d + \Psi(\eta))},$$

which converges because $s_m > d - \Psi(\eta)$. Hence there exists an almost sure event $\Omega'_{\eta,m}$ such that

$$\sum_{n=1}^{\infty} e^{-ns_m} N_n^\eta(\omega) < \infty \quad \text{for every } \omega \in \Omega'_{\eta,m}.$$

Now work on the full-probability event $\Omega_\eta \cap \Omega'_{\eta,m}$. For every $\omega \in \Omega_\eta \cap \Omega'_{\eta,m}$, the limsup covering established above and the summability of $\sum_n e^{-ns_m} N_n^\eta(\omega)$ imply $\mathcal{H}^{s_m}(E_\eta(\omega)) = 0$. Hence

$$\dim_H(E_\eta(\omega)) \leq s_m \quad \text{for every } \omega \in \Omega_\eta \cap \Omega'_{\eta,m}.$$

Finally, intersecting the countable family of full-probability events

$$\Omega_\eta \cap \bigcap_{m \geq 1} \Omega'_{\eta,m}$$

we obtain that for every ω in this intersection and every $m \geq 1$,

$$\dim_H(E_\eta(\omega)) \leq s_m.$$

Letting $m \rightarrow \infty$ yields

$$\dim_H(E_\eta) \leq d - \Psi(\eta) \quad \text{almost surely whenever } \Psi(\eta) \leq d.$$

Combining the two cases yields

$$\dim_H(E_\eta) \leq \max\{d - \Psi(\eta), 0\} \quad \text{almost surely.}$$

Finally, let $\eta_m \downarrow 0$, say $\eta_m = 1/m$. Intersecting the full-probability events on which the corresponding bounds hold for every m , and using (20), we obtain

$$\dim_H E(c) \leq \inf_{m \geq 1} \dim_H(E_{\eta_m}) \leq \inf_{m \geq 1} \max\{d - \Psi(\eta_m), 0\}.$$

The map

$$\eta \mapsto \Psi(\eta) = \inf \left\{ \frac{1}{2} a^\top Q^{-1} a : |a - c| \leq 2\eta \right\}$$

is nonincreasing, and because the quadratic form $a \mapsto \frac{1}{2} a^\top Q^{-1} a$ is continuous, the shrinking compact sets $\{a : |a - c| \leq 2\eta\}$ imply

$$\Psi(\eta) \xrightarrow{\eta \downarrow 0} \frac{1}{2} c^\top Q^{-1} c = \frac{1}{2} I.$$

Since $I < d$ in the application to $E(c)$, we may drop the positive-part operation for all sufficiently small η , and therefore

$$\dim_H E(c) \leq d - \frac{1}{2} I.$$

This proves the proposition. $\square$

Theorem 5.6 (Exact Hausdorff dimension in the L^2 -phase on the nontriviality event). *Assume the hypotheses of Theorem 4.4, in particular $I < d$. Then on the event $\{\mu^{u*}(D) > 0\}$,*

$$\dim_H E(c) = d - \frac{1}{2} I = d - \frac{1}{2} c^\top Q^{-1} c.$$

In particular, the upper bound

$$\dim_H E(c) \leq d - \frac{1}{2} I$$

holds almost surely without further assumptions.

Proof. The lower bound on $\{\mu^{u*}(D) > 0\}$ is Proposition 5.3; the upper bound is Proposition 5.5. $\square$

6 Full subcritical regime $I < 2d$

The proof of Theorem 4.4 relied on a genuine second moment computation, hence on the restriction $I < d$. In the full subcritical range $I < 2d$ one can no longer close the argument in $\mathbb{L}^2$. In this section we prove, still within the present star-scale invariant model and using the quoted scalar subcritical GMC input from Proposition 6.1, that the same limiting chaos direction governs the joint high-excursion content for *all* $I < 2d$, with convergence in probability.

6.1 A moment input for subcritical GMC

We will use a standard subcritical moment estimate for the scalar GMC in direction u_* . Since our contribution lies in the joint thick-point content theorem rather than in re-proving scalar GMC theory, we record it as a quoted input. For the exact-scaling and star-scale invariant setting considered here, the required p -moment bounds and $\mathbb{L}^1$ convergence are classical; see for instance [1, 2, 3, 5].

Proposition 6.1 (Subcritical p -moments and $\mathbb{L}^1$ convergence (quoted input)). *Let $u \in \mathbb{R}^m$ and set $I_u := u^\top \Sigma u$. Assume $I_u < 2d$. Fix any*

$$p \in \left(1, \min \left\{2, \frac{2d}{I_u}\right\}\right).$$

Then there exists a constant $C_{p,u} < \infty$ such that the following hold.

- (i) (*p -moment bound on cubes.*) *For every $t \geq 0$ and every cube $Q \subset D$ with side length $r \in (0, 1]$,*

$$\mathbb{E} \left[\mu_t^u(Q)^p \right] \leq C_{p,u} r^{\xi_u(p)}, \quad \xi_u(p) := dp - \frac{I_u}{2} p(p-1). \quad (21)$$

In particular $\sup_{t \geq 0} \mathbb{E}[\mu_t^u(D)^p] < \infty$.

- (ii) ($\mathbb{L}^1$ convergence against bounded continuous test functions.) *There exists a non-trivial random Borel measure μ^u on D such that for every bounded continuous $f : D \rightarrow \mathbb{R}$,*

$$\int_D f \, d\mu_t^u \xrightarrow[t \rightarrow \infty]{\mathbb{L}^1} \int_D f \, d\mu^u \quad \text{and} \quad \int_D f \, d\mu_t^u \xrightarrow[t \rightarrow \infty]{a.s.} \int_D f \, d\mu^u.$$

6.2 Exact conditional expectation of the thick-point content

Fix $t > s > 0$ and write $\tau := t - s$. Let $\mathcal{F}_s$ be the σ -field generated by the white noise restricted to scales $r \in [e^{-s}, 1]$. In particular, the coarse field $X_s(\cdot)$, hence also $Y_s(\cdot) = TX_s(\cdot)$ and the random measure μ_s^{u*} , are $\mathcal{F}_s$ -measurable. Then the increment field $X_t - X_s$ is independent of $\mathcal{F}_s$ and, for each fixed $x \in D$,

$$Y_t(x) - Y_s(x) \sim \mathcal{N}(0, \tau Q) \quad \text{independently of } \mathcal{F}_s.$$

For a bounded measurable test function $f : D \rightarrow \mathbb{R}$ we abbreviate

$$\nu_t(f) := \int_D f(x) \nu_t(dx), \quad \mu_s^{u*}(f e^{-\lambda \cdot g}) := \int_D f(x) e^{-\lambda \cdot g(x)} \mu_s^{u*}(dx).$$

Lemma 6.2 (Conditional Fubini). *Assume the underlying probability space is standard Borel. Let $(E, \mathcal{E}, \lambda)$ be a finite measure space, let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -field, and let $H : E \times \Omega \rightarrow \mathbb{R}$ be either*

- *bounded and $\mathcal{E} \otimes \mathcal{F}$ -measurable, or*
- *nonnegative and $\mathcal{E} \otimes \mathcal{F}$ -measurable.*

Then there exists a jointly measurable version of

$$(x, \omega) \mapsto \mathbb{E}[H(x, \cdot) \mid \mathcal{G}](\omega),$$

and for any such jointly measurable version we have

$$\mathbb{E} \left[\int_E H(x, \omega) \lambda(dx) \mid \mathcal{G} \right] = \int_E \mathbb{E}[H(x, \cdot) \mid \mathcal{G}](\omega) \lambda(dx) \quad a.s.$$

where both sides are understood in $[0, \infty]$ in the nonnegative case.

Proof. First assume that H is bounded. Since the underlying probability space is standard Borel, there exists a regular conditional probability

$$\omega \mapsto \mathbb{P}_\omega(\cdot) := \mathbb{P}(\cdot \mid \mathcal{G})(\omega).$$

Define

$$\tilde{H}(x, \omega) := \int_\Omega H(x, \omega') \mathbb{P}_\omega(d\omega').$$

Because H is bounded and jointly measurable and $\omega \mapsto \mathbb{P}_\omega$ is a probability kernel, the map $(x, \omega) \mapsto \tilde{H}(x, \omega)$ is jointly measurable. Moreover, for each fixed x , $\tilde{H}(x, \cdot)$ is a version of $\mathbb{E}[H(x, \cdot) \mid \mathcal{G}]$.

Now set

$$K(\omega) := \int_E H(x, \omega) \lambda(dx), \quad L(\omega) := \int_E \tilde{H}(x, \omega) \lambda(dx).$$

Since H is bounded and jointly measurable, both K and L are integrable, and L is $\mathcal{G}$ -measurable.

To prove that $L = \mathbb{E}[K \mid \mathcal{G}]$, it suffices to show that for every bounded $\mathcal{G}$ -measurable random variable ξ ,

$$\mathbb{E}[\xi K] = \mathbb{E}[\xi L].$$

By Fubini's theorem and the defining property of the regular conditional probability,

$$\begin{aligned} \mathbb{E}[\xi K] &= \mathbb{E} \left[\xi \int_E H(x, \omega) \lambda(dx) \right] = \int_E \mathbb{E}[\xi H(x, \cdot)] \lambda(dx) \\ &= \int_E \mathbb{E} \left[\xi(\omega) \int_\Omega H(x, \omega') \mathbb{P}_\omega(d\omega') \right] \lambda(dx) \\ &= \int_E \mathbb{E}[\xi \tilde{H}(x, \cdot)] \lambda(dx) = \mathbb{E} \left[\xi \int_E \tilde{H}(x, \omega) \lambda(dx) \right] \\ &= \mathbb{E}[\xi L]. \end{aligned}$$

Hence L is indeed a version of $\mathbb{E}[K \mid \mathcal{G}]$.

Now assume that $H \geq 0$ is merely measurable. For $m \in \mathbb{N}$, define the bounded truncations

$$H_m(x, \omega) := \min\{H(x, \omega), m\}.$$

Applying the bounded case to H_m yields

$$\mathbb{E}\left[\int_E H_m(x, \omega) \lambda(dx) \mid \mathcal{G}\right] = \int_E \mathbb{E}[H_m(x, \cdot) \mid \mathcal{G}](\omega) \lambda(dx) \quad \text{a.s.}$$

As $m \rightarrow \infty$, both sides increase monotonically:

$$H_m \uparrow H, \quad \int_E H_m d\lambda \uparrow \int_E H d\lambda.$$

Choose the conditional expectations $\mathbb{E}[H_m(x, \cdot) \mid \mathcal{G}]$ by the regular-conditional-probability formula above. Then for each x ,

$$\mathbb{E}[H_m(x, \cdot) \mid \mathcal{G}] \uparrow \mathbb{E}[H(x, \cdot) \mid \mathcal{G}] \quad \text{a.s.,}$$

and the jointly measurable representatives are increasing in m . By monotone convergence on both sides, we obtain the desired identity for nonnegative H . $\square$

Lemma 6.3 (Conditional expectation formula). *For every bounded measurable f and every $0 < s < t$ we have*

$$\mathbb{E}[\nu_t(f) \mid \mathcal{F}_s] = C_{c,Q} \left(\frac{t}{\tau}\right)^{k/2} \int_D f(x) e^{-\lambda \cdot g(x)} J_\tau(a_s(x)) \mu_s^{u_*}(dx), \quad (22)$$

where

$$a_s(x) := g(x) - (Y_s(x) - cs) \in \mathbb{R}^k$$

and $J_\tau(\cdot)$ is the deterministic factor from Lemma 4.3:

$$J_\tau(a) := \frac{\int_{[0,\infty)^k} e^{-\lambda \cdot y} \exp\left(-\frac{(a+y)^\top Q^{-1}(a+y)}{2\tau}\right) dy}{\int_{[0,\infty)^k} e^{-\lambda \cdot y} dy}. \quad (23)$$

Proof. Define

$$G(x, \omega) := C_{c,Q} \left(\frac{t}{\tau}\right)^{k/2} f(x) e^{-\lambda \cdot g(x)} J_\tau(a_s(x, \omega)) e^{u_* \cdot X_s(x, \omega) - Is/2}.$$

Since $Y_s(\cdot)$ and $X_s(\cdot)$ are jointly measurable on $D \times \Omega$, the map $(x, \omega) \mapsto a_s(x, \omega)$ is $\mathcal{B}(D) \otimes \mathcal{F}_s$ -measurable. Because $J_\tau : \mathbb{R}^k \rightarrow (0, 1]$ is Borel measurable and f, g are deterministic, G is $\mathcal{B}(D) \otimes \mathcal{F}_s$ -measurable. Moreover, $J_\tau \leq 1$ and f, g are bounded, so

$$|G(x, \omega)| \leq C |f(x)| e^{u_* \cdot X_s(x, \omega) - Is/2},$$

hence

$$\mathbb{E}\left[\int_D |G(x, \omega)| dx\right] \leq C \|f\|_\infty |D| < \infty$$

because $\mathbb{E}[e^{u_* \cdot X_s(x) - Is/2}] = 1$. In particular $\int_D G(x, \omega) dx$ is integrable and $\mathcal{F}_s$ -measurable.

We now verify that $\int_D G(x, \omega) dx$ is a version of $\mathbb{E}[\nu_t(f) \mid \mathcal{F}_s]$. Let ξ be any bounded $\mathcal{F}_s$ -measurable random variable. By ordinary Fubini,

$$\mathbb{E}[\xi \nu_t(f)] = t^{k/2} e^{It/2} \int_D f(x) \mathbb{E}[\xi \mathbf{1}_{\{Y_t(x) \geq ct + g(x)\}}] dx.$$

Fix $x \in D$. Since

$$Y_t(x) = Y_s(x) + (Y_t(x) - Y_s(x)),$$

where $a_s(x) = g(x) - (Y_s(x) - cs)$ is $\mathcal{F}_s$ -measurable and $Y_t(x) - Y_s(x) \sim \mathcal{N}(0, \tau Q)$ is independent of $\mathcal{F}_s$, we have

$$\mathbb{E}[\mathbf{1}_{\{Y_t(x) \geq ct+g(x)\}} \mid \mathcal{F}_s] = \mathbb{P}(Z_\tau \geq c\tau + a_s(x)), \quad Z_\tau \sim \mathcal{N}(0, \tau Q).$$

Therefore

$$\mathbb{E}[\xi \mathbf{1}_{\{Y_t(x) \geq ct+g(x)\}}] = \mathbb{E}[\xi \mathbb{P}(Z_\tau \geq c\tau + a_s(x))].$$

Applying Lemma 4.3 at time τ ,

$$\mathbb{P}(Z_\tau \geq c\tau + a_s(x)) = C_{c,Q} \tau^{-k/2} e^{-I\tau/2} e^{-\lambda \cdot a_s(x)} J_\tau(a_s(x)).$$

Using

$$e^{-\lambda \cdot a_s(x)} = e^{-\lambda \cdot g(x)} e^{\lambda \cdot (Y_s(x) - cs)},$$

together with $It/2 - I\tau/2 = Is/2$ and $\lambda \cdot Y_s = u_* \cdot X_s$, we obtain

$$t^{k/2} e^{It/2} \mathbb{P}(Z_\tau \geq c\tau + a_s(x)) = C_{c,Q} \left(\frac{t}{\tau}\right)^{k/2} e^{-\lambda \cdot g(x)} J_\tau(a_s(x)) e^{u_* \cdot X_s(x) - Is/2}.$$

Thus

$$t^{k/2} e^{It/2} f(x) \mathbb{E}[\xi \mathbf{1}_{\{Y_t(x) \geq ct+g(x)\}}] = \mathbb{E}[\xi G(x, \omega)].$$

Integrating over $x \in D$ and using Fubini again gives

$$\mathbb{E}[\xi \nu_t(f)] = \mathbb{E}\left[\xi \int_D G(x, \omega) dx\right].$$

Since ξ was an arbitrary bounded $\mathcal{F}_s$ -measurable random variable, this proves that $\int_D G(x, \omega) dx$ is a version of $\mathbb{E}[\nu_t(f) \mid \mathcal{F}_s]$, i.e.

$$\mathbb{E}[\nu_t(f) \mid \mathcal{F}_s] = C_{c,Q} \left(\frac{t}{\tau}\right)^{k/2} \int_D f(x) e^{-\lambda \cdot g(x)} J_\tau(a_s(x)) \mu_s^{u_*}(dx),$$

which is exactly (22). □

6.3 Controlling the J_τ -error on average

We next show that $J_\tau(a_s(x))$ may be replaced by 1 in (22) at negligible cost provided $s = o(t)$.

Lemma 6.4 (A deterministic bound on $1 - J_\tau$). *There exists a constant $C < \infty$ (depending only on Q and λ) such that for every $\tau > 0$ and every $a \in \mathbb{R}^k$,*

$$0 < J_\tau(a) \leq 1, \quad 1 - J_\tau(a) \leq \frac{C}{\tau} (1 + |a|^2). \quad (24)$$

Proof. From (23) we immediately have $0 < J_\tau(a) \leq 1$. For the second bound, use $1 - e^{-u} \leq u$ for $u \geq 0$:

$$\begin{aligned} 1 - J_\tau(a) &= \frac{\int_{[0,\infty)^k} e^{-\lambda \cdot y} \left(1 - \exp\left(-\frac{1}{2}\tau^{-1}(a+y)^\top Q^{-1}(a+y)\right)\right) dy}{\int_{[0,\infty)^k} e^{-\lambda \cdot y} dy} \\ &\leq \frac{1}{2\tau} \frac{\int_{[0,\infty)^k} e^{-\lambda \cdot y} (a+y)^\top Q^{-1}(a+y) dy}{\int_{[0,\infty)^k} e^{-\lambda \cdot y} dy}. \end{aligned}$$

Let Y be a random vector on $[0,\infty)^k$ with density proportional to $e^{-\lambda \cdot y}$. Then the ratio above is $\mathbb{E}[(a+Y)^\top Q^{-1}(a+Y)]$. Expanding and using Q^{-1} -operator norm bounds gives $\mathbb{E}[(a+Y)^\top Q^{-1}(a+Y)] \leq C(1+|a|^2)$. This proves (24). $\square$

Lemma 6.5 (Quadratic deviation under the GMC weight). *There exists $C < \infty$ such that for every $s > 0$,*

$$\mathbb{E}\left[\int_D (1 + |a_s(x)|^2) \mu_s^{u_*}(dx)\right] \leq C(1+s). \quad (25)$$

Proof. Since g is bounded and $a_s(x) = g(x) - (Y_s(x) - cs)$,

$$1 + |a_s(x)|^2 \leq C(1 + |Y_s(x) - cs|^2),$$

with C depending only on $\|g\|_\infty$. Thus it suffices to bound $\mathbb{E}\left[\int_D |Y_s(x) - cs|^2 \mu_s^{u_*}(dx)\right]$.

By Fubini and the definition of $\mu_s^{u_*}$,

$$\mathbb{E}\left[\int_D |Y_s(x) - cs|^2 \mu_s^{u_*}(dx)\right] = \int_D \mathbb{E}\left[|Y_s(x) - cs|^2 e^{u_* \cdot X_s(x) - Is/2}\right] dx.$$

For a fixed x , $X_s(x) \sim \mathcal{N}(0, s\Sigma)$, hence $Y_s(x) = TX_s(x) \sim \mathcal{N}(0, sQ)$. Moreover $u_* \cdot X_s = \lambda \cdot Y_s$ and $Q\lambda = c$, so the exponential weight

$$\exp(\lambda \cdot Y_s - \frac{1}{2}s\lambda^\top Q\lambda) = \exp(\lambda \cdot Y_s - Is/2)$$

is the Gaussian tilt that shifts the mean of Y_s from 0 to cs while preserving covariance sQ . To justify this rigorously for the unbounded function $v \mapsto |v - cs|^2$, define

$$F_M(v) := |v - cs|^2 \wedge M, \quad M \geq 1.$$

Applying Lemma 4.7 to the jointly Gaussian pair $(V, Z) = (Y_s(x), \lambda \cdot Y_s(x))$ with the bounded measurable function F_M gives

$$\mathbb{E}\left[F_M(Y_s(x)) e^{\lambda \cdot Y_s(x) - Is/2}\right] = \mathbb{E}[F_M(cs + G)], \quad G \sim \mathcal{N}(0, sQ).$$

Since $F_M \uparrow |\cdot - cs|^2$ and the Gaussian second moment is finite, monotone convergence on both sides yields

$$\mathbb{E}\left[|Y_s(x) - cs|^2 e^{\lambda \cdot Y_s(x) - Is/2}\right] = \mathbb{E}[|G|^2], \quad G \sim \mathcal{N}(0, sQ).$$

The latter equals $\text{tr}(sQ) \leq Cs$. Integrating over $x \in D$ (whose Lebesgue measure is 1) gives the claim. $\square$

Lemma 6.6 (Replacing J_τ by 1 in conditional expectation). *Let f be bounded and set $\tau = t - s$. Assume in addition that*

$$0 < s < \frac{t}{2}.$$

Then

$$\mathbb{E} \left[\left| \mathbb{E}[\nu_t(f) \mid \mathcal{F}_s] - C_{c,Q} \mu_s^{u*}(f e^{-\lambda \cdot g}) \right| \right] \leq C \|f\|_\infty \frac{1+s}{t-s}, \quad (26)$$

where $C < \infty$ depends only on (Q, λ, g) (not on s, t, f).

Proof. Subtract the expression in Lemma 6.3 from $C_{c,Q} \mu_s^{u*}(f e^{-\lambda \cdot g})$:

$$\mathbb{E}[\nu_t(f) \mid \mathcal{F}_s] - C_{c,Q} \mu_s^{u*}(f e^{-\lambda \cdot g}) = C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \left[\left(\frac{t}{\tau} \right)^{k/2} J_\tau(a_s(x)) - 1 \right] \mu_s^{u*}(dx).$$

Take absolute values and use $\mu_s^{u*} \geq 0$ and $\|f e^{-\lambda \cdot g}\|_\infty \leq \|f\|_\infty e^{\|\lambda\|_1 \|g\|_\infty}$:

$$\left| \mathbb{E}[\nu_t(f) \mid \mathcal{F}_s] - C_{c,Q} \mu_s^{u*}(f e^{-\lambda \cdot g}) \right| \leq C \|f\|_\infty \int_D \left| \left(\frac{t}{\tau} \right)^{k/2} J_\tau(a_s(x)) - 1 \right| \mu_s^{u*}(dx).$$

Now split

$$\left| \left(\frac{t}{\tau} \right)^{k/2} J_\tau(a_s) - 1 \right| \leq \left| \left(\frac{t}{\tau} \right)^{k/2} - 1 \right| + \left(\frac{t}{\tau} \right)^{k/2} |J_\tau(a_s) - 1|.$$

Since $s < t/2$, we have $\tau = t - s > t/2$ and therefore

$$1 \leq \frac{t}{\tau} < 2.$$

Set

$$u := \frac{s}{t-s} \in (0, 1), \quad \frac{t}{\tau} = 1 + u.$$

By the mean value theorem applied to $x \mapsto (1+x)^{k/2}$ on the interval $[0, 1]$, there exists a constant $C_k < \infty$ such that

$$\left| \left(\frac{t}{\tau} \right)^{k/2} - 1 \right| = |(1+u)^{k/2} - 1| \leq C_k u = C_k \frac{s}{t-s}.$$

Also, because $t/\tau < 2$, we have

$$\left(\frac{t}{\tau} \right)^{k/2} \leq 2^{k/2}.$$

Lemma 6.4 gives

$$|J_\tau(a_s) - 1| \leq \frac{C(1 + |a_s|^2)}{\tau},$$

hence

$$\left| \left(\frac{t}{\tau} \right)^{k/2} J_\tau(a_s) - 1 \right| \leq C \frac{s}{t-s} + C \frac{1 + |a_s|^2}{t-s}.$$

Integrate against μ_s^{u*} and take expectations. Since $\mathbb{E}[\mu_s^{u*}(D)] = |D| < \infty$ and Lemma 6.5 yields

$$\mathbb{E} \left[\int_D (1 + |a_s|^2) \mu_s^{u*}(dx) \right] \leq C(1 + s),$$

we obtain

$$\mathbb{E} \left[\int_D \left| \left(\frac{t}{\tau} \right)^{k/2} J_\tau(a_s) - 1 \right| \mu_s^{u*}(dx) \right] \leq C \frac{s}{t-s} + C \frac{1+s}{t-s} \leq C \frac{1+s}{t-s}.$$

This proves (26). $\square$

6.4 A mesoscopic decoupling and a p -moment fluctuation estimate

We now show that $\nu_t(f)$ is close to its coarse conditional expectation $\mathbb{E}[\nu_t(f) \mid \mathcal{F}_s]$ in $\mathbb{L}^p$ for some $p > 1$ by a spatial decoupling at a mesoscopic scale. The argument uses only: (i) finite range of correlations in the increment field $X_t - X_s$ (coming from the compact support of ϕ), (ii) the subcritical p -moment bound of Proposition 6.1, and (iii) the elementary exponential domination $\mathbf{1}_{\{Y_t \geq ct+g\}} \leq e^{\lambda \cdot (Y_t - ct - g)}$.

Choice of parameters. Assume from now on that $I = I_{u_*} < 2d$. Pick

$$p \in \left(1, \min \left\{2, \frac{2d}{I}\right\}\right). \quad (27)$$

Then $\xi_{u_*}(p) > d$ by (21). Fix any $A > 0$ such that

$$A(\xi_{u_*}(p) - d) > \frac{kp}{2}. \quad (28)$$

For each $t \geq e$ define the mesoscopic time

$$s = s(t) := A \log t, \quad r = r(t) := e^{-s} = t^{-A}. \quad (29)$$

Then $s(t) \rightarrow \infty$ and $s(t) = o(t)$ as $t \rightarrow \infty$.

Cube partition and coloring. Partition $D = [0, 1]^d$ into the regular grid of half-open cubes of mesh r , denoted by $\mathcal{Q}_r$; cubes meeting the boundary may be smaller, but all have side length at most r and the total number of cubes is $\asymp r^{-d}$. There exists a deterministic partition (a “ 3^d -coloring”)

$$\mathcal{Q}_r = \bigsqcup_{\ell=1}^{3^d} \mathcal{Q}_r^{(\ell)}$$

such that whenever $Q, Q' \in \mathcal{Q}_r^{(\ell)}$ with $Q \neq Q'$, we have $\text{dist}(Q, Q') \geq r$.

For each $Q \in \mathcal{Q}_r$ define the cube contribution

$$Z_{t,Q} := \int_Q f(x) t^{k/2} e^{It/2} \mathbf{1}_{\{Y_t(x) \geq ct+g(x)\}} dx, \quad m_{t,Q} := \mathbb{E}[Z_{t,Q} \mid \mathcal{F}_s], \quad \delta_{t,Q} := Z_{t,Q} - m_{t,Q}. \quad (30)$$

Then

$$\nu_t(f) = \sum_{Q \in \mathcal{Q}_r} Z_{t,Q}, \quad \nu_t(f) - \mathbb{E}[\nu_t(f) \mid \mathcal{F}_s] = \sum_{Q \in \mathcal{Q}_r} \delta_{t,Q}.$$

Lemma 6.7 (Conditional independence of mesoscopic blocks). *Fix $t > s$ and let $r = e^{-s}$. For each color class ℓ , the family $\{\delta_{t,Q} : Q \in \mathcal{Q}_r^{(\ell)}\}$ is conditionally independent given $\mathcal{F}_s$, and each $\delta_{t,Q}$ is centered: $\mathbb{E}[\delta_{t,Q} \mid \mathcal{F}_s] = 0$.*

Proof. Centering is immediate from the definition.

For each cube $Q \in \mathcal{Q}_r$, let

$$Q^+ := \{z \in \mathbb{R}^d : \text{dist}(z, Q) \leq r/4\},$$

and let $\mathcal{G}_Q$ be the σ -field generated by the white noise restricted to

$$Q^+ \times [e^{-t}, e^{-s}).$$

We claim that $Z_{t,Q}$ is $\sigma(\mathcal{F}_s, \mathcal{G}_Q)$ -measurable.

Indeed, $Y_s(x)$ is $\mathcal{F}_s$ -measurable for every $x \in D$, while the increment field $X_t - X_s$ is obtained by integrating the white noise over scales $r' \in [e^{-t}, e^{-s})$. Because ϕ is supported in $B(0, 1/4)$, the value $(X_t - X_s)(x)$ only depends on the white noise in the set

$$\{(z, r') : r' \in [e^{-t}, e^{-s}), |z - x| \leq r'/4\}.$$

If $x \in Q$, then $r' \leq e^{-s} = r$, so $|z - x| \leq r'/4 \leq r/4$ implies $z \in Q^+$. Hence the whole collection $\{(X_t - X_s)(x) : x \in Q\}$ is $\mathcal{G}_Q$ -measurable. Since

$$Y_t(x) = Y_s(x) + T(X_t - X_s)(x),$$

it follows that the event $\{Y_t(x) \geq ct + g(x)\}$ is $\sigma(\mathcal{F}_s, \mathcal{G}_Q)$ -measurable for every $x \in Q$, and therefore the integral $Z_{t,Q}$ is $\sigma(\mathcal{F}_s, \mathcal{G}_Q)$ -measurable. Consequently

$$m_{t,Q} = \mathbb{E}[Z_{t,Q} \mid \mathcal{F}_s]$$

is $\mathcal{F}_s$ -measurable, and hence $\delta_{t,Q} = Z_{t,Q} - m_{t,Q}$ is also $\sigma(\mathcal{F}_s, \mathcal{G}_Q)$ -measurable.

Now fix a color class ℓ and distinct cubes $Q, Q' \in \mathcal{Q}_r^{(\ell)}$. By construction $\text{dist}(Q, Q') \geq r$, so

$$\text{dist}(Q^+, (Q')^+) \geq r - \frac{r}{4} - \frac{r}{4} = \frac{r}{2} > 0.$$

Therefore the space-scale sets

$$Q^+ \times [e^{-t}, e^{-s}) \quad \text{and} \quad (Q')^+ \times [e^{-t}, e^{-s})$$

are disjoint. Since the white noise is independently scattered, $\mathcal{G}_Q$ and $\mathcal{G}_{Q'}$ are independent and, more generally, the family $(\mathcal{G}_Q)_{Q \in \mathcal{Q}_r^{(\ell)}}$ is independent.

To conclude conditional independence, it is enough to prove that the sigma-fields

$$\mathcal{H}_j := \sigma(\mathcal{F}_s, \mathcal{G}_{Q_j}), \quad j = 1, \dots, m,$$

are conditionally independent given $\mathcal{F}_s$, because each δ_{t,Q_j} is $\mathcal{H}_j$ -measurable.

For each j , let $\mathcal{A}_j$ be the algebra of bounded random variables generated by products

$$\xi(\omega) \mathbf{1}_{A_j}(\omega), \quad \xi \in L^\infty(\mathcal{F}_s), \quad A_j \in \mathcal{G}_{Q_j}.$$

Then $\sigma(\mathcal{A}_j) = \mathcal{H}_j$.

We first verify the conditional product identity on $\mathcal{A}_1 \times \dots \times \mathcal{A}_m$. By multilinearity it is enough to consider

$$U_j = \xi_j \mathbf{1}_{A_j}, \quad \xi_j \in L^\infty(\mathcal{F}_s), \quad A_j \in \mathcal{G}_{Q_j}.$$

Since $\prod_{j=1}^m \xi_j$ is $\mathcal{F}_s$ -measurable and the sigma-fields $\mathcal{G}_{Q_j}$ are independent and independent of $\mathcal{F}_s$, for every bounded $\mathcal{F}_s$ -measurable random variable ζ we have

$$\mathbb{E}\left[\zeta \prod_{j=1}^m U_j\right] = \mathbb{E}\left[\zeta \prod_{j=1}^m \xi_j \prod_{j=1}^m \mathbf{1}_{A_j}\right] = \mathbb{E}\left[\zeta \prod_{j=1}^m \xi_j\right] \prod_{j=1}^m \mathbb{P}(A_j).$$

On the other hand,

$$\mathbb{E}[U_j \mid \mathcal{F}_s] = \xi_j \mathbb{P}(A_j),$$

so

$$\mathbb{E}\left[\zeta \prod_{j=1}^m \mathbb{E}[U_j \mid \mathcal{F}_s]\right] = \mathbb{E}\left[\zeta \prod_{j=1}^m \xi_j\right] \prod_{j=1}^m \mathbb{P}(A_j).$$

Since this holds for every bounded $\mathcal{F}_s$ -measurable ζ , we obtain

$$\mathbb{E}\left[\prod_{j=1}^m U_j \mid \mathcal{F}_s\right] = \prod_{j=1}^m \mathbb{E}[U_j \mid \mathcal{F}_s] \quad \text{a.s.}$$

for all $U_j \in \mathcal{A}_j$.

We now upgrade from $\mathcal{A}_j$ to all bounded $\mathcal{H}_j$ -measurable random variables by induction on j . Suppose that for some $r \in \{0, \dots, m\}$ we already know the identity whenever

$$U_1 \in L^\infty(\mathcal{H}_1), \dots, U_r \in L^\infty(\mathcal{H}_r), \quad U_{r+1} \in \mathcal{A}_{r+1}, \dots, U_m \in \mathcal{A}_m.$$

(For $r = 0$, this is exactly the algebra case proved above.)

Fix $U_1, \dots, U_r \in L^\infty(\mathcal{H}_1), \dots, L^\infty(\mathcal{H}_r)$ and $U_{r+2} \in \mathcal{A}_{r+2}, \dots, U_m \in \mathcal{A}_m$, and define $\mathcal{D}_{r+1}$ to be the class of bounded $\mathcal{H}_{r+1}$ -measurable random variables V such that

$$\mathbb{E}\left[U_1 \cdots U_r V U_{r+2} \cdots U_m \mid \mathcal{F}_s\right] = \mathbb{E}[V \mid \mathcal{F}_s] \prod_{j \neq r+1} \mathbb{E}[U_j \mid \mathcal{F}_s] \quad \text{a.s.}$$

By the induction hypothesis, $\mathcal{A}_{r+1} \subset \mathcal{D}_{r+1}$. Moreover $\mathcal{D}_{r+1}$ is a vector space and is closed under bounded monotone convergence (because conditional expectation is linear and monotone). Since $\mathcal{A}_{r+1}$ is an algebra generating $\mathcal{H}_{r+1}$, the functional monotone class theorem implies

$$\mathcal{D}_{r+1} = L^\infty(\mathcal{H}_{r+1}).$$

Thus the conditional product identity holds when U_{r+1} is an arbitrary bounded $\mathcal{H}_{r+1}$ -measurable random variable. This closes the induction step.

Iterating from $r = 0$ to $r = m$, we conclude that for all bounded $\mathcal{H}_j$ -measurable random variables U_j ,

$$\mathbb{E}\left[\prod_{j=1}^m U_j \mid \mathcal{F}_s\right] = \prod_{j=1}^m \mathbb{E}[U_j \mid \mathcal{F}_s] \quad \text{a.s.}$$

Applying this with

$$U_j = F_j(\delta_{t, Q_j}),$$

where $F_j : \mathbb{R} \rightarrow \mathbb{R}$ are bounded Borel functions, yields

$$\mathbb{E}\left[\prod_{j=1}^m F_j(\delta_{t, Q_j}) \mid \mathcal{F}_s\right] = \prod_{j=1}^m \mathbb{E}[F_j(\delta_{t, Q_j}) \mid \mathcal{F}_s] \quad \text{a.s.}$$

This is exactly the conditional independence of the family $\{\delta_{t, Q} : Q \in \mathcal{Q}_r^{(\ell)}\}$ given $\mathcal{F}_s$. $\square$

Lemma 6.8 (Conditional von Bahr–Esseen inequality). *Assume the underlying probability space is standard Borel. Let $\mathcal{G}$ be a sub- σ -field, let $p \in (1, 2]$, and let $\xi_1, \dots, \xi_N \in L^p$ be random variables that are conditionally independent and conditionally centered given $\mathcal{G}$. Then*

$$\mathbb{E} \left[\left| \sum_{j=1}^N \xi_j \right|^p \mid \mathcal{G} \right] \leq 2 \sum_{j=1}^N \mathbb{E} [|\xi_j|^p \mid \mathcal{G}] \quad a.s.$$

Proof. Since the underlying probability space is standard Borel, there exists a regular conditional probability kernel

$$\omega \longmapsto \mathbb{P}_\omega(\cdot) := \mathbb{P}(\cdot \mid \mathcal{G})(\omega).$$

Choose a countable π -system $\mathcal{P}$ generating $\mathcal{B}(\mathbb{R})$, for instance the family of half-lines

$$(-\infty, q], \quad q \in \mathbb{Q}.$$

Because conditional independence means that for every choice $A_j \in \mathcal{P}$,

$$\mathbb{E} \left[\prod_{j=1}^N \mathbf{1}_{\{\xi_j \in A_j\}} \mid \mathcal{G} \right] = \prod_{j=1}^N \mathbb{E} [\mathbf{1}_{\{\xi_j \in A_j\}} \mid \mathcal{G}] \quad a.s.,$$

and because there are only countably many such tuples $(A_1, \dots, A_N)$, we may fix a $\mathcal{G}$ -measurable event Ω_0 of full probability on which all these identities hold simultaneously.

Since $\mathbb{E}[\xi_j \mid \mathcal{G}] = 0$ a.s. and $\xi_j \in L^p \subset L^1$, after shrinking Ω_0 if necessary we may also assume that for every $\omega \in \Omega_0$ and every j ,

$$\int_{\Omega} \xi_j(\omega') \mathbb{P}_\omega(d\omega') = 0, \quad \int_{\Omega} |\xi_j(\omega')|^p \mathbb{P}_\omega(d\omega') = \mathbb{E}[|\xi_j|^p \mid \mathcal{G}](\omega) < \infty.$$

Fix $\omega \in \Omega_0$. For every $A_j \in \mathcal{P}$, the defining property of the regular conditional probability gives

$$\mathbb{P}_\omega(\xi_1 \in A_1, \dots, \xi_N \in A_N) = \int \prod_{j=1}^N \mathbf{1}_{\{\xi_j \in A_j\}} d\mathbb{P}_\omega = \mathbb{E} \left[\prod_{j=1}^N \mathbf{1}_{\{\xi_j \in A_j\}} \mid \mathcal{G} \right](\omega),$$

while similarly

$$\mathbb{P}_\omega(\xi_j \in A_j) = \mathbb{E}[\mathbf{1}_{\{\xi_j \in A_j\}} \mid \mathcal{G}](\omega).$$

Hence for every $\omega \in \Omega_0$ and every $A_j \in \mathcal{P}$,

$$\mathbb{P}_\omega(\xi_1 \in A_1, \dots, \xi_N \in A_N) = \prod_{j=1}^N \mathbb{P}_\omega(\xi_j \in A_j).$$

Since $\mathcal{P}$ is a π -system generating $\mathcal{B}(\mathbb{R})$, the π - λ theorem implies that $\xi_1, \dots, \xi_N$ are independent under $\mathbb{P}_\omega$ for every $\omega \in \Omega_0$. Moreover they are centered and belong to $L^p(\mathbb{P}_\omega)$.

We may therefore apply the classical von Bahr–Esseen inequality under $\mathbb{P}_\omega$ to obtain, for every $\omega \in \Omega_0$,

$$\mathbb{E}_{\mathbb{P}_\omega} \left[\left| \sum_{j=1}^N \xi_j \right|^p \right] \leq 2 \sum_{j=1}^N \mathbb{E}_{\mathbb{P}_\omega} [|\xi_j|^p].$$

By the definition of the regular conditional expectation, this is exactly

$$\mathbb{E}\left[\left|\sum_{j=1}^N \xi_j\right|^p \mid \mathcal{G}\right](\omega) \leq 2 \sum_{j=1}^N \mathbb{E}[|\xi_j|^p \mid \mathcal{G}](\omega),$$

for every $\omega \in \Omega_0$, proving the claim. $\square$

Lemma 6.9 (*p*-moment bound on the fluctuation). *With the choices (27)–(29), we have*

$$\mathbb{E}\left[\left|\nu_t(f) - \mathbb{E}[\nu_t(f) \mid \mathcal{F}_{s(t)}]\right|^p\right] \xrightarrow{t \rightarrow \infty} 0. \quad (31)$$

Proof. Write $s = s(t)$, $r = e^{-s}$ for brevity.

We first verify that each $\delta_{t,Q}$ belongs to L^p . Since $\delta_{t,Q} = Z_{t,Q} - m_{t,Q}$ with $m_{t,Q} = \mathbb{E}[Z_{t,Q} \mid \mathcal{F}_s]$, Jensen gives

$$|m_{t,Q}|^p \leq \mathbb{E}[|Z_{t,Q}|^p \mid \mathcal{F}_s] \quad \text{a.s.}$$

Hence

$$|\delta_{t,Q}|^p \leq 2^{p-1}(|Z_{t,Q}|^p + |m_{t,Q}|^p) \leq 2^{p-1}(|Z_{t,Q}|^p + \mathbb{E}[|Z_{t,Q}|^p \mid \mathcal{F}_s]).$$

Taking expectations, we obtain

$$\mathbb{E}[|\delta_{t,Q}|^p] \leq 2^p \mathbb{E}[|Z_{t,Q}|^p].$$

So it suffices to prove that $Z_{t,Q} \in L^p$, and to bound its p -th moment.

We now bound $|Z_{t,Q}|$ by the corresponding chaos mass. Since $\lambda > 0$, on the event $\{Y_t(x) \geq ct + g(x)\}$ we have $\lambda \cdot (Y_t(x) - ct - g(x)) \geq 0$, hence

$$\mathbf{1}_{\{Y_t(x) \geq ct + g(x)\}} \leq \exp(\lambda \cdot (Y_t(x) - ct - g(x))).$$

Therefore, using $u_* \cdot X_t = \lambda \cdot Y_t$ and $\|f\|_\infty < \infty$,

$$\begin{aligned} |Z_{t,Q}| &\leq \|f\|_\infty t^{k/2} e^{It/2} \int_Q \exp(\lambda \cdot (Y_t - ct - g)) \, dx \\ &= \|f\|_\infty t^{k/2} \int_Q e^{-\lambda \cdot g(x)} \exp(u_* \cdot X_t(x) - It/2) \, dx \\ &\leq C \|f\|_\infty t^{k/2} \mu_t^{u_*}(Q), \end{aligned}$$

where $C = \sup_{x \in D} e^{-\lambda \cdot g(x)} < \infty$. Thus

$$\mathbb{E}[|Z_{t,Q}|^p] \leq C \|f\|_\infty^p t^{kp/2} \mathbb{E}[\mu_t^{u_*}(Q)^p]. \quad (32)$$

Apply Proposition 6.1 to the cube Q . If $\ell(Q)$ denotes its side length, then $\ell(Q) \leq r = t^{-A}$. Since $\xi_{u_*}(p) > d > 0$, we have $\ell(Q)^{\xi_{u_*}(p)} \leq r^{\xi_{u_*}(p)}$, and therefore

$$\mathbb{E}[\mu_t^{u_*}(Q)^p] \leq C_{p,u_*} \ell(Q)^{\xi_{u_*}(p)} \leq C_{p,u_*} r^{\xi_{u_*}(p)}.$$

Hence every $\delta_{t,Q} \in L^p$, and the conditional von Bahr–Esseen inequality may be applied.

By Lemma 6.7, for each ℓ the variables $\{\delta_{t,Q}\}_{Q \in \mathcal{Q}_r^{(\ell)}}$ are conditionally independent and conditionally centered given $\mathcal{F}_s$. Since $p \in (1, 2]$, Lemma 6.8 gives

$$\mathbb{E}\left[\left|\sum_{Q \in \mathcal{Q}_r^{(\ell)}} \delta_{t,Q}\right|^p \mid \mathcal{F}_s\right] \leq 2 \sum_{Q \in \mathcal{Q}_r^{(\ell)}} \mathbb{E}[|\delta_{t,Q}|^p \mid \mathcal{F}_s].$$

Summing over ℓ and using the elementary inequality $|\sum_{\ell=1}^{3^d} a_\ell|^p \leq (3^d)^{p-1} \sum_{\ell=1}^{3^d} |a_\ell|^p$ gives

$$\begin{aligned} \mathbb{E}\left[\left|\sum_{Q \in \mathcal{Q}_r} \delta_{t,Q}\right|^p \mid \mathcal{F}_s\right] &\leq C \sum_{\ell=1}^{3^d} \mathbb{E}\left[\left|\sum_{Q \in \mathcal{Q}_r^{(\ell)}} \delta_{t,Q}\right|^p \mid \mathcal{F}_s\right] \\ &\leq C \sum_{Q \in \mathcal{Q}_r} \mathbb{E}[|\delta_{t,Q}|^p \mid \mathcal{F}_s]. \end{aligned}$$

Taking expectations and using the bound on $|\delta_{t,Q}|^p$ above together with Jensen yields

$$\mathbb{E}\left[|\nu_t(f) - \mathbb{E}[\nu_t(f) \mid \mathcal{F}_s]|^p\right] \leq C \sum_{Q \in \mathcal{Q}_r} \mathbb{E}[|Z_{t,Q}|^p]. \quad (33)$$

The number of cubes is $|\mathcal{Q}_r| \asymp r^{-d}$, so inserting (32) yields

$$\mathbb{E}\left[|\nu_t(f) - \mathbb{E}[\nu_t(f) \mid \mathcal{F}_s]|^p\right] \leq C \|f\|_\infty^p t^{kp/2} r^{-d} r^{\xi_{u_*}(p)} = C \|f\|_\infty^p t^{kp/2} r^{\xi_{u_*}(p)-d}.$$

With $r = t^{-A}$ this becomes

$$\mathbb{E}\left[|\nu_t(f) - \mathbb{E}[\nu_t(f) \mid \mathcal{F}_s]|^p\right] \leq C \|f\|_\infty^p t^{kp/2-A(\xi_{u_*}(p)-d)}.$$

By our choice of A in (28), the exponent is negative, hence the right-hand side tends to 0 as $t \rightarrow \infty$. This proves (31). $\square$

6.5 Convergence in probability for all $I < 2d$

Theorem 6.10 (Joint thick-point content in the full subcritical regime). *Assume Q is positive definite and $\lambda = Q^{-1}c \in (0, \infty)^k$. Let*

$$u_* := T^\top Q^{-1}c, \quad I := \|u_*\|_\Sigma^2 = c^\top Q^{-1}c,$$

and assume the full subcritical condition $I < 2d$. Let μ^{u_} denote the subcritical GMC limit furnished by Proposition 6.1(ii). Then for every bounded continuous $f : D \rightarrow \mathbb{R}$,*

$$\nu_t(f) \xrightarrow[t \rightarrow \infty]{P} C_{c,Q} \int_D f(x) e^{-\lambda \cdot g(x)} \mu^{u_*}(dx).$$

Consequently,

$$\nu_t \xrightarrow[t \rightarrow \infty]{P} C_{c,Q} e^{-\lambda \cdot g} \mu^{u_*} \quad \text{in the topology of weak convergence of finite measures on } D. \quad (34)$$

Proof. Fix a bounded continuous f . Choose p and A as in (27)–(28) and set $s = s(t)$ as in (29). Write $\tau = t - s$.

Decompose the error into three terms:

$$\begin{aligned} \left| \nu_t(f) - C_{c,Q} \mu^{u_*}(f e^{-\lambda \cdot g}) \right| &\leq \underbrace{\left| \nu_t(f) - \mathbb{E}[\nu_t(f) \mid \mathcal{F}_s] \right|}_{(1)} \\ &\quad + \underbrace{\left| \mathbb{E}[\nu_t(f) \mid \mathcal{F}_s] - C_{c,Q} \mu_s^{u_*}(f e^{-\lambda \cdot g}) \right|}_{(2)} \\ &\quad + \underbrace{\left| C_{c,Q} \left[\mu_s^{u_*}(f e^{-\lambda \cdot g}) - \mu^{u_*}(f e^{-\lambda \cdot g}) \right] \right|}_{(3)}. \end{aligned}$$

Term (1). Lemma 6.9 gives $(1) \rightarrow 0$ in $\mathbb{L}^p$, hence in probability.

Term (2). Since $s(t) = A \log t = o(t)$, there exists $t_0 = t_0(A)$ such that

$$0 < s(t) < \frac{t}{2} \quad \text{for all } t \geq t_0.$$

Hence Lemma 6.6 applies for all sufficiently large t , and gives

$$\mathbb{E}[(2)] \leq C \|f\|_\infty \frac{1 + s(t)}{t - s(t)} \xrightarrow{t \rightarrow \infty} 0.$$

Therefore $(2) \rightarrow 0$ in $\mathbb{L}^1$, hence in probability.

Term (3). Apply Proposition 6.1(ii) with the bounded continuous test function $\varphi(x) := f(x)e^{-\lambda \cdot g(x)}$. Since g is bounded continuous, so is φ , and therefore

$$\int_D \varphi(x) \mu_s^{u*}(\mathrm{d}x) \xrightarrow{s \rightarrow \infty} \int_D \varphi(x) \mu^{u*}(\mathrm{d}x).$$

Because $s(t) \rightarrow \infty$, this implies $(3) \rightarrow 0$ in $\mathbb{L}^1$, hence in probability.

Putting the three terms together yields

$$\nu_t(f) \rightarrow C_{c,Q} \mu^{u*}(f e^{-\lambda \cdot g}) \quad \text{in probability}$$

for every bounded continuous f . To conclude weak convergence in probability of the measures, we again argue along arbitrary deterministic sequences. Let $t_n \rightarrow \infty$. The convergence just proved for bounded continuous test functions implies that, for every such f ,

$$\nu_{t_n}(f) \xrightarrow[n \rightarrow \infty]{P} C_{c,Q} \mu^{u*}(f e^{-\lambda \cdot g}).$$

Applying Lemma 4.8 to the sequence $(\nu_{t_n})_{n \geq 1}$ gives

$$\nu_{t_n} \xrightarrow[n \rightarrow \infty]{P} C_{c,Q} e^{-\lambda \cdot g} \mu^{u*} \quad \text{weakly as finite measures.}$$

Applying Lemma 4.9 in the metric space of finite Borel measures on D equipped with the bounded-Lipschitz metric, we conclude that the family $(\nu_t)_{t \geq 0}$ satisfies (34). $\square$

6.6 Remarks

Remark 6.11 (What was used beyond $\mathbb{L}^2$?). Relative to the $\mathbb{L}^2$ -phase proof, the only genuinely new ingredients are:

- a mesoscopic decomposition $t = s + \tau$ with $s = s(t) \uparrow \infty$ but $s = o(t)$, so that the exact one-point tail identity yields a good approximation of the conditional expectation, and
- a spatial block decoupling at scale e^{-s} combined with a subcritical p -moment estimate for the chaos μ_t^{u*} , used to control the fluctuation $\nu_t(f) - \mathbb{E}[\nu_t(f) \mid \mathcal{F}_s]$ in $\mathbb{L}^p$ for some $p > 1$.

No two-point asymptotics for the excursion event are needed in this step; the entire non- $\mathbb{L}^2$ difficulty is absorbed into the moment estimate for the limiting GMC.

7 Transfer issues and links with circle models and circular ensembles

Theorems 4.4 and 6.10 were proved in the explicit compactly supported star-scale invariant model of Section 2. This choice was deliberate: it provides exact scale decomposition and makes the genuinely joint probabilistic estimates fully checkable line by line, while standard scalar GMC input is kept separate and explicitly quoted.

At the same time, the form of the limiting expression is not tied to the particular star-scale construction used in the proof. It depends only on the logarithmic covariance matrix Σ , the linear constraint matrix T , and the resulting minimum-energy direction u_* . This makes it reasonable to investigate analogous statements for more general log-correlated Gaussian fields

$$\text{Cov}(X(x), X(y)) = \Sigma \log \frac{1}{|x - y|} + G(x, y),$$

provided one can realize the field, locally on compact sets, as a sum of:

- a star-scale invariant component with the same logarithmic singularity,
- an independent continuous perturbation,
- and a remainder that vanishes uniformly under the regularization scale.

Versions of such decompositions are available in several scalar settings in the literature, together with uniqueness results for the associated GMC; see for instance [3, 5, 6]. In a vector-valued setting, however, what would be needed here is a *simultaneous* decomposition that works coherently for all components and preserves the covariance matrix structure. We do not claim such a theorem here, and no transfer result is used elsewhere in the paper; formulating and verifying assumptions under which an appropriate simultaneous decomposition holds is left for future work.

A second natural direction concerns circle models and random matrices. In the subcritical regime, recent work of Lambert–Najnudel [9] establishes subcritical GMC limits for circular β -ensemble observables, including the characteristic polynomial and the counting field. This motivates, rather than proves, the possibility that the joint thick-point content problem studied here may admit a random-matrix counterpart, at least in suitable mesoscopic regularizations.

The probabilistic message emerging from our results is the following: when several correlated logarithmic observables are simultaneously forced to be atypically large, the resulting content measure is governed, in the explicit Gaussian model treated here, by the unique minimum-energy direction solving the corresponding linear constraints. Extending this principle to broader classes of log-correlated fields and to concrete random-matrix observables requires additional input and remains an important next step.

8 Discussion and outlook

The perspective of the paper is that simultaneous thick-point geometry should be treated as a *measure-valued* problem and not only as a dimension problem. The minimum-energy direction u_* encodes the effective observable selected by the joint constraint and is therefore the correct object to compare across models.

Two directions seem especially natural, but at present they should be viewed as directions rather than as theorem-level consequences of the present paper. First, one would like to sharpen the general-kernel discussion above into a directly verifiable theorem for broad classes of log-correlated kernels, without passing through an abstract simultaneous decomposition statement. Second, in the random-matrix direction, one would like to identify a concrete finite- n regime in which the current Gaussian model can be matched to characteristic-polynomial or counting-field observables with the precision required for a measure-valued thick-point limit. From the probabilistic viewpoint, these questions belong to the general programme of understanding how several correlated log-observables interact at the level of high points, overlaps, and induced chaos measures.

9 Bibliographic note on scalar subcritical GMC input

In Section 6.1 we use Proposition 6.1 as quoted scalar GMC input. For the exact-scaling and star-scale invariant kernels considered here, the finiteness of positive moments below the Seiberg threshold, the corresponding power-law mass estimates for balls, and the $\mathbb{L}^1$ convergence of the martingale approximation are standard consequences of the classical scalar theory; see Kahane [1], Robert–Vargas [2], Rhodes–Vargas [3], and Shamov [5]. The cube version stated in Proposition 6.1 is then obtained by a routine ball/cube comparison in the bounded domain D . We therefore treat Proposition 6.1 as quoted background and do not reproduce its proof here.

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